On constraining initial orbital eccentricity of purely inspiral events

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The LIGO-Virgo-KAGRA consortium has sporadically detected purely inspiral gravitational-wave events like GW170817 and GW190425. These events offer an opportunity to constrain the presence of possible initial (residual) orbital eccentricities while using purely inspiral template families. We detail the implementation of a LSC Algorithm Library Suite approximant, `TaylorF2Ecck`, which models analytically inspiral gravitational waves from non-spinning compact binaries in Post-Newtonian accurate eccentric orbits while restricting the initial-eccentricity contributions to its leading order. Specifically, our frequency domain approximant consistently incorporates the orbital, advance of periastron, and gravitational-wave emission effects fully up to the 3PN order. We pursue detailed parameter estimation studies of GW170817 and GW190425 using `TaylorF2Ecck`, after performing comprehensive sanity checks to validate the model's performance and investigate the influence of periastron advance in the relevant parameter space. The results indicate that initial-eccentricity at 20 Hz is essentially zero (less than 0.011 for GW170817 and less than 0.028 for GW190425) at a 90% confidence level. We compare posteriors for these two events that arise from a few purely inspiral eccentric approximants and point out certain subtle effects arising due to the inclusion of periastron advance. Additionally, these detailed studies reveal the importance of incorporating initial eccentricity contributions at least up to 3.5PN-order and discuss its implications. We substantiate this inference by employing versions of quasi-circular `TaylorF2` approximant that incorporate Fourier phase contributions beyond the conventional 3.5PN order.

I. INTRODUCTION

The LIGO-Virgo-KAGRA (LVK) collaboration has been routinely detecting transient gravitational-wave (GW) events from the coalescence of stellar-mass black hole (BH) binaries during their observing runs [? ? ? ? ? ?]. With the detection of over 100 such events, the era of GW astronomy has been firmly established [? ? ? ?]. Additionally, the consortium had sporadically detected GWs from a few neutron star-neutron star (NS-NS or BNS) and neutron star-black hole (NS-BH) binaries [? ?]. The discovery of inspiral GWs from merging NS-NS binary in GW170817, along with its electromagnetic counterparts, marked the advent of multi-messenger GW astronomy [?]. It is important to note that the currently operational GW observatories are primarily sensitive to the inspiral phase of GWs from binary NS systems [?].

Such inspiral-dominated GW events are typically detected using the fully analytical frequency-domain `TaylorF2` approximant within the LSC Algorithm Library Suite (LALSuite) [?]. This approximant employs the Post-Newtonian (PN) approximation to General Relativity to model the temporally evolving GW polarization states associated with the inspiral phase of compact binary coalescence [?]. Note that the PN approximation requires slow motion and weak fields, and hence requires $(v/c)^2 \ll 1$ and $(Gm/rc^2) \ll 1$, where v, m, and r are the orbital velocity, total mass, and relative separation of the binary, respectively. The computationally efficient and fully analytical frequency-domain `TaylorF2` approximant models inspiral GWs from compact binaries in quasi-circular orbits. The standard LALSuite implementation typically employs Newtonian (quadrupolar) order amplitude corrections, while the phase corrections are accurate to 3.5PN order, providing $(v/c)^7$ order corrections to the dominant quadrupolar contributions [?]. This approximant utilizes the stationary phase approximation (SPA) to obtain analytical frequency-domain inspiral templates from their time-domain counterparts [?]. It is possible to extend the approach to incorporate the effects of spin and orbital eccentricity [? ? ? ? ? ? ? ?].

Purely inspiral GW events, such as GW170817, spent substantially longer durations in the LVK frequency window (around 20 Hz to kHz) compared to typical BH binary events. For instance, the inspiral phase of GW170817 spans approximately 168[1] s, while that of the first GW event, GW150914, lasted a mere 0.7?? s. Consequently, purely inspiral LVK events are particularly well-suited for exploring the effects of any residual orbital eccentricity, benefiting from their large number of GW cycles within the operational frequency windows of GW observatories [?]. In general, typical LVK binaries are expected to exhibit small residual eccentricities,

[1] Inspiral duration estimated using the `calculate_time_to_merger` function in `Bilby` software [?], from 20 Hz to the last stable orbit frequency $f_{\rm LSO}$ (see Eq. (??) of Appendix ??).

less than 10^{-4}, although NS binaries may display higher initial eccentricities [?]. The small initial eccentricities are expected due to the efficiency of GWs in circularizing isolated eccentric binaries by rapidly radiating away angular momentum [?]. Astrophysical considerations suggest that the detection of residual eccentricities in LVK events is crucial for distinguishing between the two possible astrophysical formation channels for stellar-mass compact binaries [?]. In the isolated formation channel, compact binaries originate from the evolution of massive binary stars without external perturbation [?], and astrophysical assessments indicate that their orbits should be circularized by the time their GWs enter the LVK frequency window [?]. In contrast, compact binaries can evolve dynamically when BHs or NSs form binaries with other BHs or NSs in dense stellar environments [?]. Such systems include nuclear star clusters, globular clusters, young star clusters, or even the discs of Active Galactic Nuclei (for a detailed review, see Refs. [? ? ? ? ? ? ? ?]). It is argued that the presence of measurable initial orbital eccentricity should aid in distinguishing between these scenarios [? ? ? ? ? ? ? ? ? ? ?]. This understanding was the primary motivation behind early attempts to constrain the initial eccentricity of GW150914, as discussed in Ref. [?]. Significant work has gone into creating gravitational waveform models that incorporate the effects of orbital eccentricity[? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ?]. Such models have been applied to analyze GW signals observed by LIGO and Virgo, and they have also been used in recent studies to set bounds on initial eccentricities of a few LVK coalescence events[? ? ?].

Interestingly, firm bounds on initial-eccentricity e_0 of two pure inspiral LVK events, namely GW170817 and GW190425, were established in Ref. [?]. This analysis utilized the `TaylorF2ecc` approximant, which incorporates leading-order initial eccentricity contributions that are 3PN-accurate in the Fourier phase of the widely used circular `TaylorF2` inspiral approximant [?]. The `TaylorF2ecc` approximant, as detailed in Ref. [? ?], also includes 3.5PN-order circular inspiral contributions to the dominant Fourier phase component and features quadrupolar and circular amplitudes. However, it does not account for the effects of periastron advance which enter the compact binary dynamics at the 1PN order [?]. This effect refers to the gradual rotation of the closest point (periastron) of an eccentric orbit involving compact objects due to general relativistic gravitational effects.

Our current work introduces a new LALSuite-compatible approximant, `TaylorF2ecck`, which utilizes and adapts inputs from Ref. [?]. Consequently, our effort provides fully analytic Fourier-domain inspiral templates for eccentric inspirals, incorporating $\mathcal{O}(e_0^2)$ contributions arising from 3PN-accurate periastron advance and GW emission in the relevant Fourier phases, while their amplitudes are at the quadrupolar order, accurate to $\mathcal{O}(e_0)$ order. In other words, our inspiral approximant extends the `TaylorF2` inspiral approximant for non-spinning compact binaries by consistently including all leading-order e_0 contributions that arise from orbital, periastron advance, and GW emission effects—with these contributions included in the phase up to 3PN order and in the amplitude at the 0PN order. Therefore, our `TaylorF2ecck` approximant provides an accurate and computationally efficient and accurate template family for analyzing non-spining low-eccentricity inspiral events. We note in passing that this is a restricted version of the approximant presented in Ref. [?], which provides eccentric corrections accurate to $\mathcal{O}(e_0^6)$ with amplitude corrections at 1PN order.

In this paper, we employ our `TaylorF2ecck` approximant to constrain e_0 of two pure inspiral GW events, GW170817 and GW190425. By applying Bayesian inference techniques and nested sampling algorithms [?], we estimate e_0 for these events, yielding results consistent with near-zero eccentricity around a GW frequency of 20 Hz. at a high confidence level. Interestingly, our analysis shows no significant difference in the results between uniform and logarithmic priors though Ref. [?] had pointed out a dependence on the choice of prior for constraining e_0 while employing another eccentric inspiral `TaylorF2ecc` approximant. Our study reveals the subtle influences of the periastron effect, particularly in the higher mass region of the parameter space as this conservative PN effect mainly depends on the total mass. Additionally, these detailed studies strongly suggest the importance of incorporating initial eccentricity contributions at least up to the 3.5PN-order, influenced by the nature of posteriors for the mass ratio for the pure inspiral GW events. We substantiate this inference by employing versions of the quasi-circular `TaylorF2` approximant that incorporate Fourier phase contributions beyond the conventional 3.5PN order.

This manuscript is structured as follows: Sec. ?? provides an overview of the various frequency-domain waveforms employed in our investigations. In this section, the TaylorF2ecck approximant is introduced, and waveform sanity checks are performed. Sec. ?? details the data analysis techniques and presents parameter estimation results for GW170817 and GW190425. Our efforts to constrain the e_0 value for these events, along with its implications and limitations, are described within this section. Appendices ?? and ?? offer the explicit PN-accurate expressions relevant to our inspiral approximants.

II. INSPIRAL WAVEFORM APPROXIMANTS

In this section, we briefly introduce various fully analytic inspiral approximants employed in our investigations. These include existing approximants available in the LALSimulation library, such as `TaylorF2` and `TaylorF2ecc`, which are detailed in Refs. [? ?], and two new eccentric inspiral approximants: `TaylorF2ecck` and `TaylorF2ecch` influenced by [? ? ?]. We first describe these approximants, followed by presenting detailed sanity checks for them.

A. Eccentric Inspiral Approximants

We begin by listing our `TaylorF2ecck` approximant that employs expressions from Ref. [?]. This fully analytic Fourier domain approximant models inspiral GWs from non-spinning compact point mass binaries in preccessing PN-accurate eccentric orbits. While Ref. [?] offers expressions with 1PN-accurate amplitude corrections and 3PN-accurate phase evolution, both incorporating eccentricity effects up to $\mathcal{O}(e_0^6)$ at each PN order, our implementation for this work adopts a restricted version, primarily to reduce computation cost in parameter estimation. Specially, for the present effort, we restrict our attention to a version that incorporates the quadrupolar order amplitudes, while the relevant Fourier phases are fully 3PN-accurate to the leading order $\mathcal{O}(e_0^2)$ contributions, and $\mathcal{O}(e_0)$ contributions in the Fourier amplitude. Our `TaylorF2ecck` approximant adapted from Eq. (2.5) of Ref. [?] reads

$$
\begin{aligned}
\tilde{h}(f) =& \tilde{\mathcal{A}} \left(\frac{Gm\pi f}{c^3}\right)^{-7/6} \\
& \times \sum_{j,n} \xi_{j,n} \left(\frac{j}{2}\right)^{2/3} e^{-i(\Psi_{j,n}-\pi/4)}\, \Theta_{j,n}\,, \qquad (1a)
\end{aligned}
$$

with

$$
\tilde{\mathcal{A}} = -\left(\frac{5\pi\eta}{384}\right)^{1/2} \frac{(Gm)^2}{c^5 D_L}\,, \qquad (1b)
$$

where (j,n) harmonic combinations take values (1,0), (1,-2), (2,-2) and (3,-2) for the current version of the `TaylorF2ecck` approximant. The (2,-2) harmonic is the dominant one while the rest are sub-dominant, with their amplitudes suppressed at low e_0 values (see Eq. (??)). Following Ref. [?], we term $\tilde{A}$ and $\xi_{j,n}$ as harmonic independent and harmonic dependent amplitudes of our approximant, respectively. Clearly, $\tilde{A}$ depends on the total mass ($m = m_1 + m_2$), the symmetric-mass-ratio ($\eta = \frac{m1\, m2}{m^2} = \frac{q}{(1+q)^2}$), where m_1, m_2 and q are the individual masses and their ratio and D_L stands for the luminosity-distance. Note that it is possible to express the m and η dependence of $\tilde{A}$ into the more familiar chirp-mass $\mathcal{M}$ dependence [?]. In the above equation, $\Theta_{j,n}$ is the Unit-Step function, which allows us to appropriately terminate each harmonic at the correct physically allowed frequency limit (see Eq. (3.26) of Ref. [?]). Furthermore, $\xi_{j,n}$ depends on e_0, orbital inclination ι, and azimuthal angles β, along with the antenna pattern functions F_+, $F_\times$, and η (we let $\beta = 0$, influenced by the `LALSimulation` considerations). In what follows, we list the expressions for $\xi_{j,n}$ that are accurate to $\mathcal{O}(e_0)$ order, influenced by Eq. (3.16) in Ref. [?], and it reads

$$
\begin{aligned}
\xi_{1,0} &= F_+ \left\{-\frac{\sin^2\iota}{\chi^{19/18}} e_0\right\}, \\
\xi_{1,-2} &= F_+ \left\{-\frac{3(3+\cos(2\iota))}{4\chi^{19/18}} e_0\right\} + iF_\times \left\{-\frac{\cos\iota}{\chi^{19/18}} e_0\right\}, \\
\xi_{2,-2} &= F_+ \left\{3+\cos(2\iota)\right\} + iF_\times \left\{4\cos\iota\right\}, \\
\xi_{3,-2} &= F_+ \left\{\frac{9(3+\cos(2\iota))}{4\chi^{19/18}} e_0\right\} + iF_\times \left\{\frac{9\cos\iota}{\chi^{19/18}} e_0\right\}, \qquad (2)
\end{aligned}
$$

where the dimensionless χ is given by $\chi = f/f_0$ (with f_0 being the the initial GW frequency). Note that the dominant (2,-2) mode lack e_0 correction and circular in nature.

In what follows, we list fully analytic 1PN-accurate Fourier phases, namely (j,n) harmonics, that include all the $\mathcal{O}(e_0^2)$ corrections for visualization purposes. This form for the (j,n) harmonic reads

$$
\begin{aligned}
\Psi_{j,n} =& -2\pi f t_c + \phi_c\left(j-(j+n)\frac{k}{1+k}\right) - \frac{3j}{256\eta x_{j,n}^{5/2}}\Bigg[1 - \frac{2355 e_0^2}{1462\chi^{19/9}} + x_{j,n}\Bigg\{\frac{55\eta}{9} - \frac{25n}{3j} - \frac{2585}{756} \\
& + e_0^2\left(\frac{\frac{154645\eta}{17544} - \frac{2223905}{491232}}{\chi^{25/9}} + \frac{-\frac{128365\eta}{12432} + \frac{1805n}{172j} + \frac{69114725}{14968128}}{\chi^{19/9}}\right)\Bigg\} + \ldots\Bigg], \qquad (3a)
\end{aligned}
$$

with

$$
x_{j,n} = \left[\frac{2\pi G m f}{\left\{j-(j+n)\frac{k}{1+k}\right\} c^3}\right]^{2/3}, \qquad (3b)
$$

and we refer to Eq. (??) of Appendix ?? for its 3PN-accurate version. In the above equation, t_c and ϕ_c are the time and phase of coalescence, respectively, while k is the rate of periastron advance and $x_{j,n}$ is the harmonic-dependent dimensionless PN parameter. We would like to emphasize that our *restricted* `TaylorF2ecck` approximant contains *four* harmonics, while their Fourier phases are fully 3PN-accurate and incorporate $\mathcal{O}(e_0^2)$ corrections. For the present effort, we employ the 3PN-accurate version of k in our approximant, given in Eq. (??) of Appendix ??.

To isolate the effects of k, we now introduce another eccentric approximant, namely `TaylorF2ecch`, influenced by Refs. [? ?]. Therefore, this approximant omits the effects of periastron advance both on Fourier amplitudes and phases though Fourier phases are 3PN-accurate while incorporating $\mathcal{O}(e_0^2)$ corrections. The neglect of periastron advance effects on amplitudes ensures that the `TaylorF2ecch` is expressed as a sum over single harmonics j, whereas TaylorF2ecck includes splitting (as seen in the splitting of $j = 1$ to $(j, n) = (1, 0), (1, -2)$, as evident from Eq. (??)). In other words, this approximant has a reduced harmonic content with modes limited to $j = 1, 2, 3$ and lacks the additional phase shift associated with periastron effects found in the sub-dominant harmonics of TaylorF2ecck (see the second term in Eq. (??) for $\psi_{j,n}$). Therefore, this approximant should help us delineate scenarios where periastron effects are significant in comparison with our `TaylorF2ecck` templates, as discussed in Sec. ??). Our `TaylorF2ecch` approximant, influenced by Ref. [?], reads

$$\tilde{h}(f) = \tilde{\mathcal{A}} \left(\frac{Gm\pi f}{c^3} \right)^{-7/6} \times \sum_j \xi_j \left(\frac{j}{2} \right)^{2/3} e^{-i(\Psi_j - \pi/4)} \, \Theta_j \,, \tag{4}$$

while its Fourier amplitudes have the following structure

$$\begin{aligned} \xi_1 =& F_+ \left\{ -\frac{3(3+\cos(2\iota))}{4\chi^{19/18}} e_0 - \frac{\sin(\iota)^2}{\chi^{19/18}} e_0 \right\} \\ &+ iF_\times \left\{ -\frac{\cos(\iota)}{\chi^{19/18}} e_0 \right\} , \\ \xi_2 =& F_+ \left\{ 3 + \cos(2\iota) \right\} + iF_\times \left\{ 4 \, \cos(\iota) \right\} , \\ \xi_3 =& F_+ \left\{ \frac{9(3+\cos(2\iota))}{4\chi^{19/18}} e_0 \right\} + iF_\times \left\{ \frac{9 \, \cos(\iota)}{\chi^{19/18}} e_0 \right\} . \end{aligned} \tag{5}$$

The 3PN-accurate expression of Ψ_j is included in Appendix ?? (see Eq. (??)). For the sake of comparison with Eq. (??), the 1PN-accurate version of Fourier phases is given by

$$\begin{aligned} \Psi_j =& -2\pi f t_c + \phi_c j - \frac{3j}{256\eta x^{5/2}} \Bigg[1 + \frac{2355 e_0^2}{1462 \chi^{19/9}} \\ &+ x_j \Bigg\{ \frac{55\eta}{9} + \frac{3715}{756} + \mathrm{et0}^2 \Bigg(\frac{-\frac{128365\eta}{12432} - \frac{2045665}{348096}}{\chi^{19/9}} \\ &+ \frac{\frac{154645\eta}{17544} - \frac{2223905}{491232}}{\chi^{25/9}} \Bigg) \Bigg\} + \dots \Bigg] . \end{aligned} \tag{6}$$

We now describe a version of the `TaylorF2ecc` approximant, detailed in Sec. IV of Ref. [?] and mentioned in Sec. II of Ref. [?], that we employed in our present effort. This approximant essentially extends the `TaylorF2` approximant by incorporating leading-order e_0 corrections to its PN-accurate Fourier phase. This implies that the amplitude of this approximant is circular and accurate to the Newtonian (quadrupolar) order. Consequently, its harmonic content is restricted to the dominant (2,-2) mode [?]. As detailed in Ref. [?], `TaylorF2ecc` also includes 3.5PN-order circular contributions to its Fourier phase. To make meaningful comparisons in our analysis, we typically ignore circular non-spinning 3.5PN terms and PN-accurate (circular) spin contributions that appear in the Fourier phase expression of the `TaylorF2ecc` approximant, unless stated otherwise. Let us emphasize that `TaylorF2ecc` does not account for the effects of periastron advance both in its Fourier amplitudes and phases as evident from Eq. (6.26) of Ref. [?].

Model	Fourier Phase accuracy	Amplitude correction	Harmonics (j, n)
`TaylorF2ecck`	3PN-$\mathcal{O}(e_0^2)$	$\mathcal{O}(e_0)$	(1, 0), (1, -2), (2, -2), (3, -2)
`TaylorF2ecch`	3PN-$\mathcal{O}(e_0^2)$	$\mathcal{O}(e_0)$	(1, -2), (2, 2), (3, -2)
`TaylorF2ecc`	3PN-$\mathcal{O}(e_0^2)$	None	(2, -2)
`TaylorF2`	3, 3.5, 4, 4.5PN	None	(2, -2)

TABLE I. This table compares inspiral approximants used in the current study. All approximants consider non-spinning compact binaries with amplitudes at quadrupolar order. `TaylorF2ecck` features 3PN-$\mathcal{O}(e_0^2)$ accurate Fourier Phases, multiple harmonics beyond the (2,-2) mode, quadrupolar amplitude with $\mathcal{O}(e_0)$ corrections, all effects of periastron advance. `TaylorF2ecch` is similar to `TaylorF2ecck` in phasing and amplitude accuracy (PN and e_0 order) but omits all periastron advance related effects. `TaylorF2ecc` has 3PN-$\mathcal{O}(e_0^2)$ accurate Fourier phase corrections but is restricted to the dominant harmonic with quadrupolar, circular amplitude, while neglecting periastron effects [?]. Our quasi-circular `TaylorF2` template family allows us to invoke it with its Fourier phase accurate to different PN orders.

Finally, let us briefly discuss various versions of the quasi-circular `TaylorF2` approximant, detailed in Ref. [?], that we employed in the present effort. Recall that this approximant works for purely circular inspirals and employs a 3.5PN accurate Fourier phase expression, given by Eq. (3.18) in Ref. [?]. The standard LALSuite implementation of `TaylorF2` typically employs Newtonian (quadrupolar) order amplitude corrections, with phase corrections accurate to 3.5PN order with respect to the dominant quadrupolar contribution. In order to understand various systematics present in the parameter estimation studies of our eccentric approximants, we employ versions of the `TaylorF2` approximant where the Fourier phase expressions are 3PN, 3.5PN, 4PN, and 4.5PN accurate. For the last two versions, we employed Eq. (9) in Ref. [?] which provides a fully

4.5PN-accurate Fourier phase for circular inspirals. In what follows, we list in Table ?? all the waveform approximants that we employed for the present set of investigations.

B. Waveform Sanity Checks

Initial verifications of our `TaylorF2Ecck` eccentric inspiral approximants within the `LALSimulation` framework were performed by comparing it with `TaylorF2Ecch` and the well-established `TaylorF2` and `TaylorF2Ecc` inspiral template families. These checks focused on computational efficiency, waveform amplitude and phase evolution with frequency, and convergence to the quasi-circular `TaylorF2` approximant as eccentricity approaches zero. Further checks were conducted to determine the range of parameter space for which the inclusion of periastron advance becomes important in both matched-filtering searches and parameter estimation. Deviations of `TaylorF2Ecck` from `TaylorF2`, `TaylorF2Ecc`, and `TaylorF2Ecch` were examined in the m–e_0 and η–e_0 parameter spaces. All analyses in this section use a minimum frequency of 20 Hz, a choice discussed in detail in Sec. ??. Additionally, we use **GW170817-like** parameters obtained from the median values of the posterior distribution from a parameter estimation run with `TaylorF2Ecck` (see Sec. ?? and Appendix ?? for details). It should be noted that `TaylorF2Ecch` is included in these tests for context only. Its waveform behavior will not be elaborated upon in the results of Section ?? and it will not be included in the event analysis of Sec. ??.

We typically use $|\tilde{h}_+(f)|$ while probing amplitude evolution of our approximants and we have checked that the conclusions remain essentially unchanged if we employ $|\tilde{h}_\times(f)|$. In Fig. ??, we display $|\tilde{h}_+(f)|$ for `TaylorF2Ecck`, `TaylorF2Ecch`, `TaylorF2Ecc`, and `TaylorF2` approximants, using simulated GW150914 (short inspiral, BBH merger) and GW170817 (long inspiral, BNS merger) events to illustrate their behavior for both short and long inspiral scenarios and we let $e_0 = 0.1$ at 20 Hz. Refer to Eq. (??) of Appendix ?? for the analytical formulation of eccentricity evolution with frequency.

For $e_0 > 0.02$, both `TaylorF2Ecck` and `TaylorF2Ecch` approximants exhibit visible amplitude modulations due to overlapping harmonic contributions, but the morphology of this modulation is different due to `TaylorF2Ecck` having extra harmonics due to side band(s), and extra phase shift present in the sub-dominant harmonics. For low-(total) mass systems, these approximants exhibit a higher number of modulations per frequency bin due to their longer inspiral durations. These modulations diminish at higher frequencies as the eccentricity decreases and these waveforms converge towards their circular `TaylorF2` counterpart. This is mainly because these harmonics are inversely proportional to $f^{19/18}$ through the χ variable as evident from Eqs. (??). Interestingly, we infer that frequency evolution of the real part of $\tilde{h}_+(f)$ is fairly similar between our eccentric models while diverging from the quasi-circular `TaylorF2` model.

To assess the impact of eccentricity and periastron advance on both matched-filtering searches and parameter estimation, and also to facilitate a comparison with the match study presented in Ref. [?], we conduct match calculations and injection-recovery tests across a range of e_0, m and η values. The match $\mathcal{M}$ quantifies the phase difference and waveform similarity between two waveform families and it reads [?]

$$\mathcal{M}(h_s, h_t) = \max_{t_0,\phi_0} \left\{ 4\mathrm{Re} \int_{f_{\mathrm{low}}}^{f_{\mathrm{high}}} \frac{\tilde{h}_s^*(f)\tilde{h}_t(f)}{S_h(f)}\, df \right\} . \quad (7)$$

Note that the above equation maximizes the overlap integral (the inner product of the frequency-domain waveforms $\tilde{h}_s(f)$ and $\tilde{h}_t(f)$, weighted by the inverse of detector's noise power spectral density (PSD), denoted by $S_h(f)$), with respect to the coalescence time (t_0) and the associated phase (ϕ_0). For our efforts, $S_h(f)$ is given by LIGO's O4 design sensitivity PSD [?] for the L1 detector. The integration element df is considered to be $1/\Delta\mathrm{T}$ in the numerical integration, where $\Delta\mathrm{T}$ is the waveform's time-domain duration. For our analysis, we defined and used the mismatch-percentage $\mathcal{M}^*_{\%} = (1-\mathcal{M}) \times 100\%$, instead of $\mathcal{M}$ values [? ?]. A match of $\mathcal{M} = 0.97$ leads to $\mathcal{M}^*_{\%} = 3\%$ and this is generally considered acceptable for matched-filtering searches, corresponding to a signal loss of less than 10% [?].

For the injection-recovery test, we consider an idealized scenario where `TaylorF2Ecck` waveform is injected in zero-noise. We then use `TaylorF2Ecck`, `TaylorF2Ecch`, `TaylorF2Ecc` and `TaylorF2` waveform templates to recover it via Bayesian parameter estimation, employing a reduced 5-parameter space (i.e. $\mathcal{M}$, q, e_0, ϕ_c and t_c) instead of the standard 16 parameters [?] to ensure reasonable computational cost. Waveform distinguishability is assessed through differences in the resulting Bayes factors ($\Delta \log_{10} \mathrm{BF}$), with values of $1-2$ indicating strong evidence and > 2 indicating decisive distinguishability [?]. We also note that alternative (less stringent) distinguishability criteria exist, with a threshold of $\Delta \log \mathrm{BF} = 8$ often used as the level of "strong evidence" in favor of one hypothesis over another (Jeffreys, 1961) [?], which I will simply refer to as Jeffreys' criteria. This injection-recovery framework further enables us to explore how `TaylorF2Ecck` deviates from other waveform families as the injected parameters are varied. In this context, the $\Delta \log_{10} \mathrm{BF} > 2$ criterion serves as a practical proxy for identifying significant waveform differences. The analysis assumes the L1 and H1 detectors operating at their O4 design sensitivity, with corresponding PSDs. The technicalities of parameter estimation and Bayes-Factor calculations are detailed in Sec. ??.

Fig. ?? presents a comparative analysis of the `TaylorF2Ecck` approximant against other template families using the match calculations and injection-recovery tests. In Fig. ??, $\mathcal{M}^*_{\%}$ is shown as a function of e_0, where three colored bands correspond to match calculations between TaylorF2Ecck and i) TaylorF2Ecch, ii) TaylorF2Ecc, and iii) TaylorF2 approximants, respectively. Each band reflects multiple match computations with varying component masses ($m_1, m_2 \in [1M_\odot, 3M_\odot]$) within the BNS prior range used in Ref. [?]. We employ $\mathcal{M}^*_{\%}$ rather than $\mathcal{M}$ to facilitate easy comparisons of results that arise from the distinguishability metric, $\Delta \log_{10} \mathrm{BF}$.

Fig. ?? displays $\Delta \log_{10} \mathrm{BF}$ values from the injection-recovery test, where signals are generated with our TaylorF2Ecck inspiral waveform family, using the GW170817-like parameters. The relevant parameters are recovered using TaylorF2Ecck, TaylorF2Ecch, TaylorF2Ecc, and TaylorF2 inpsiral template families. Unlike match calculations, this analysis does not include multiple parameter estimation runs while varying injected mass parameters due to their higher computational costs. Injections were performed at four discrete e_0 values between 0.0 and 0.2 for each comparison. After the recovery, discrete values of $\Delta \log_{10} \mathrm{BF}$ were interpolated using cubic splines. We observe mild variations in these results when we change mass parameters appropriate for GW170817.

We now discuss our inferences from plots present in Fig. ??. The quasi-circular TaylorF2 waveform diverges rapidly, exceeding 3% mismatch at $e_0 \sim 0.015$. At low eccentricities, all eccentric waveforms exhibit similar phase evolution. The TaylorF2Ecch and TaylorF2Ecc models reach 3% mismatch at $e_0 \sim 0.16$ and $e_0 \sim 0.23$, respectively, slightly lower than the $e_0 \sim 0.25$ reported in [?] (which compared eccentric models with and without periastron advance effects), likely due to differences in the employed PSDs and the use of beyond-leading order eccentric corrections in [?]. Injection-recovery tests indicate that TaylorF2Ecck becomes distinguishable from TaylorF2Ecch and TaylorF2Ecc at $e_0 \gtrsim 0.05$ and decisively preferred at $e_0 \gtrsim 0.06$. This provides insights into the e_0 parameter space region where the periastron effect significantly influences parameter estimation. However, TaylorF2Ecc and TaylorF2Ecch successfully recovered injected parameters within 90% credible intervals even for $e_0 > 0.15$. In contrast, TaylorF2 failed to recover the injected mass parameters even at $e_0 \geq 0.01$, producing inaccurate parameter estimation posteriors for GW170817-like events with $e_0 \geq 0.01$. Furthermore, TaylorF2Ecck becomes distinguishable from TaylorF2 at $e_0 \gtrsim 0.011$, thus inferring a region in e_0 space where the eccentricity effect is expected to significantly influence parameter estimation. Additional posterior plots of these analyses are included in Appendix ??.

To examine deviations between template families and underlying correlations in relevant parameters, we conduct several match tests in e_0–m and e_0–η parameter spaces. Unless stated otherwise, the remaining parameters are set to the GW170817-like parameter values. While similar investigations with injection-recovery tests are prohibitively expensive, we could infer the rough behavior of such parameter estimation studies from our match tests. The plan is to use changing $\mathcal{M}^*_{\%}$ values as a proxy for waveform deviations and the inclined lines of non-changing $\mathcal{M}^*_{\%}$ values to indicate parameter correlations (see Fig. ??). We note that such correlations should not be confused with those arising from our detailed parameter estimation posterior plots. The behavior observed in these tests when comparing TaylorF2Ecck and TaylorF2Ecc will be revisited during the comparison of parameter estimation results for GW170817 in Sec. ??.

Fig. ?? illustrates how $\mathcal{M}^*_{\%}$ varies with changes in m, e_0, and η when comparing TaylorF2Ecck with TaylorF2Ecc, in order to probe the impact of including periastron advance and additional harmonics. An extended comparison involving TaylorF2Ecch and TaylorF2 is provided in Appendix??. When comparing TaylorF2Ecck to TaylorF2Ecc, the deviation between the models increases with both e_0 and m, accompanied by a negative correlation (see top panel of Fig. ??). A similar trend is observed in the e_0–η plane, where a weaker but still negative correlation appears (see bottom panel of Fig. ??). Moreover, the variation in $\mathcal{M}^*_{\%}$ with increasing η at fixed e_0 is notably smaller, and negligible at lower e_0, compared to the variation observed with increasing m.

To conclude, TaylorF2Ecck diverges from TaylorF2 even at small eccentricities ($e_0 \gtrsim 0.011$) during parameter estimation results, where eccentricity correction is important. The e_0 limits from Ref. [?] have a similar conclusion, indicating significant systematic bias when eccentricity is ignored, $e_0 \gtrsim 0.01$ at 10 Hz. Moreover, periastron effect becomes important at small eccentricities of $e_0 \gtrsim 0.06$. It is important to remember that these injection-recovery tests are conducted in an idealized scenario with zero noise, O4 design sensitivity (as the analysis PSD), and a reduced parameter space; conservative estimates (stricker, more realistic estimates) are expected to be higher than these values.

Following these results, we can, in general, consider posterior values of $e_0 \lesssim 0.011$ as essentially representing zero eccentricity for BNS (GW170817-like) events, since this is the regime where eccentric and circular waveform models become indistinguishable in parameter estimation, as reflected in their Bayes factors. However, it's important to note that at low SNR or at higher values of m or both, the uncertainty in e_0 will be higher [?], and the upper limit of 90% credible interval might extend beyond this value, even when the event is circular in nature. To further investigate such scenario, we repeated similar injection-recovery tests for a GW190425-like event, in a manner analogous to the analysis performed with the GW170817-like event, to compare the eccentric TaylorF2Ecck and quasi-circular TaylorF2 templates. Because of the higher mass and lower SNR associated with this event, we observed that TaylorF2Ecck becomes distinguishable from TaylorF2 at a slightly higher value of $e_0 \gtrsim 0.018$, and strongly distinguishable at $e_0 \gtrsim 0.028$ according to Jeffreys' criteria (for the extended results, refer to Fig. ?? in Appendix ??). Furthermore, TaylorF2 failed to recover the injected mass parameters for $e_0 \gtrsim 0.02$. In what follows, for parameter estimation in Sec. ??, we will consider a value e_0 as essentially zero eccentricity (or negligible eccentricity) if the 90% credible interval from parameter estimation with a uniform e_0 prior is below a conservative value of 0.011 for GW170817 and 0.028 for GW190425. We will analyze GW190425 less strictly, and also consider the result from parameter estimation with log-prior on e_0 as an additional sanity check for the presence or absence of residual eccentricity in the event. Moreover, the sanity tests in this section also indicate that, when $e_0 \neq 0$, TaylorF2Ecck should diverge noticeably from TaylorF2Ecc in parameter estimations within the e_0–m parameter spaces (similarly in m–e_0 space). This observations will be investigated in the parameter estimations of GW170817 and GW190425.

III. BAYESIAN PARAMETER ESTIMATION STUDIES ON PURE INSPIRAL EVENTS

This section presents Bayesian parameter estimation studies conducted on a couple of pure inspiral GW events. We first provide a brief overview of the Bayesian methodology, widely used within the LVK collaboration for GW data analysis [? ? ?]. This is followed by a detailed description of our parameter estimation studies on a couple of pure inspiral events, comparing our eccentric inspiral templates with existing waveform families and exploring their astrophysical and data-analysis implications.

A. Bayesian Inference Framework

Bayesian inference provides a robust framework for estimating various orbital parameters of GW events. This approach involves fitting the observed data, $d(t)$, with a chosen inspiral template family, H, under the assumption of Gaussian detector noise. The posterior probability density, $\mathcal{P}(\theta|d(t),H)$, for the source parameters θ given the data and the model is obtained via Bayes' theorem, as

$$\mathcal{P}(\theta|d(t),H) = \frac{\mathcal{L}(d(t)|\theta,H)\,\pi(\theta|H)}{\mathcal{Z}(d(t)|H)}\,, \tag{1}$$

where $\mathcal{L}(d(t)|\theta,H)$ is the likelihood function, $\pi(\theta|H)$ is the prior distribution for the parameters, and $\mathcal{Z}(d(t)|H)$ is the Bayesian evidence given by

$$\mathcal{Z}(d(t)|H) = \int \mathcal{L}(d(t)|\theta,H)\,\pi(\theta|H)\,d\theta\,. \tag{2}$$

Note that the likelihood function quantifies the agreement between the data and the model for a given set of parameters, weighted against the Gaussian noise profile. For our analysis, we employ the Bilby Python package [? ?], which implements both Nested Sampling and Markov-Chain Monte Carlo (MCMC) algorithms for posterior distribution computations. The Nested Sampling, as compared to MCMC used in Ref. [?], is particularly advantageous for model comparison in higher-dimensional parameter spaces, as it provides an estimate of the Bayesian evidence, allowing for the calculation of the Bayes factor. This factor for a given parameter estimation is given by $\mathcal{B} = \mathcal{Z}/\mathcal{Z}_{\rm noise}$, and for two parameter estimation studies with model A and B by $\mathcal{B}^A_B = \mathcal{Z}_A/\mathcal{Z}_B$. Note that $\mathcal{B}^A_B$ quantifies the relative evidence for one model over another, allowing for a robust assessment of comparative model performance in signal recovery. For waveform comparisons in our analysis, we use the metric $\Delta\log_{10}{\rm BF} = \log_{10}(\mathcal{B}^A_B)$, already employed in the injection-recovery tests in Sec. ??. In addition, we employ parameter-specific metrics, for intuitive comparison between our posterior distributions: the percentage difference in median values (%ΔMedian), and the overlap of the 90% credible intervals ($\mathcal{O}^{90\%}_{\rm CI}$). The percentage difference in median values is defined as

$$\%\Delta{\rm Median} = \frac{|{\rm Med}_1 - {\rm Med}_2|}{\frac{{\rm Med}_1+{\rm Med}_2}{2}} \times 100\%\,, \tag{3}$$

where ${\rm Med}_1$ and ${\rm Med}_2$ are the median values of a parameter from the two comparing models. The overlap of the 90% credible intervals is calculated by

$$\mathcal{O}^{90\%}_{\rm CI} = \begin{cases} 0\% & \text{, if } l_1 > u_2 \text{ or} \\ & \quad l_2 > u_1 \\ \frac{\min(u_1,u_2)-\max(l_1,l_2)}{\min(u_1-l_1,u_2-l_2)} \times 100\% & \text{, otherwise}\,, \end{cases} \tag{4}$$

where l_1 and u_1 are the lower and upper bounds of model A's 90% credible interval, and l_2 and u_2 are those from model B. Derived from fundamental statistical principles [? ?], these two additional metrics together indicate the proximity of two posterior distributions of a parameter for two comparing waveform models. Specifically, %ΔMedian quantifies the relative difference in central tendency and assesses systematic shifts, while $\mathcal{O}^{90\%}_{\rm CI}$ quantifies overlap or consistency of estimated uncertainties provided by the 90% credible interval.

Our eccentric inspiral template families are implemented in LALSimulation, which is part of the LALSuite and written primarily in C for computational efficiency. Bilby is then used to interface with LALSimulation for parameter estimation studies. Detailed settings and all parameter estimation results for this literature are stored in [?]. We ignored tidal effects for our parameter estimation runs while prior distributions are set similar to Ref. [?]. We typically let spin parameters to be zero unless stated otherwise, especially when employing templates for quasi-circular inspiral.

The choice of analysis duration and $f_{\min}$ significantly impacts both the accuracy and computational costs of our parameter estimation runs. To capture eccentricity effects, the maximum possible data length, determined by the mass-prior, should be used. For mass priors with m_1, $m_2 \in [1M_\odot, 3M_\odot]$, the maximum data length corresponds to $\sim$ 309 seconds (1.1 times the chirp time [?]) for a $1M_\odot$-$1M_\odot$ system, and it exceeds typical LVK analysis durations for BNS events (128-190 seconds) [? ?]. Although lowering $f_{\min}$ could improve eccentricity measurement[? ?], it drastically increases computational demands; for instance, decreasing $f_{\min}$ from 20 Hz to 10 Hz for GW170817 would require a sixfold increase in data duration. Furthermore, the detector noise at 10 Hz is significantly higher than at 20 Hz [? ?], and potentially reducing the SNR and introducing additional uncertainties. Given these factors, and the recommendation of Ref. [?] that 20 Hz is optimal for parameter estimation, we set $f_{\min}$ to 20 Hz, acknowledging the potential loss of information below this frequency. To manage computational cost and achieve a reasonable parameter estimation run within a week, we also set the sampling frequency to 2048 Hz and $f_{\max}$ to 256 Hz for GW170817. This choice is justified given that our primary interest lies in the inspiral phase, particularly the early inspiral where signatures of residual eccentricity, if present, are expected to be most pronounced. This approach is further supported by similar strategies in recent literature, such as the use of $f_{\max} = 280$ Hz to investigate eccentricity in the inspiral phase of a NS-BH system[?].

The PSD for each analysis is estimated from up to 1024 s of pre-event strain data using Welch's method, as implemented in `bilby_pipe` [?]. The data are divided into segments with analysis duration, 309 s for GW170817 and 160 s for GW190425, with 50% overlap, each tapered by a Tukey window with a 0.4 s roll-off. The median power spectrum across all segments is computed to yield a typical PSD estimate for parameter estimation.

B. Parameter estimation implications of `TaylorF2ecck` approximant for Pure Inspiral Events GW170817 and GW190425

We are now in a position to pursue parameter estimation studies of our eccentric approximants while employing data associated with pure inspiral events. This effort primarily aims to investigate the data analysis implications of `TaylorF2ecck` and compare them with parameter estimation results while using the `TaylorF2ecc` approximant, reported in Refs. [? ?]. Parameter estimation with the `TaylorF2Ecck` approximant poses significant computational challenges due to its complex harmonic structure and the prolonged duration of BNS signals. For instance, our Bayesian parameter estimation runs for GW170817 using `TaylorF2Ecck` took more than a week to complete, whereas the same analysis with the `TaylorF2Ecc` approximant required only half of the time. To mitigate this computational burden, we adopted the prior outlined in the Ref. [?], which utilizes electromagnetic counterpart data for GW170817 to fix the sky localization and constrain the luminosity distance prior to a Gaussian distribution about 40.7 Mpc. However, this approach is not feasible for GW190425 event, which lacks an electromagnetic counterpart and has a lower network signal-to-noise ratio (SNR) of $\sim$13 compared to GW170817's SNR of $\sim$33. This lower SNR complicates parameter constraints for GW190425, yet its inclusion in our study demonstrates that key findings from GW170817 extend to other BNS inspiral events.

While a detailed analysis of spin effects is beyond the scope of this paper, Fig. ?? in Appendix?? offers a broader context. It compares posteriors from our 3PN-accurate `TaylorF2ecck` approximant with those from: (i) the standard full-`TaylorF2ecc` approximant (incorporating leading-order spin effects and 3.5PN circular contributions to its Fourier phase) using our parameter estimation results; (ii) the same full-`TaylorF2ecc` model analyzed by Lenon et al. 2020 (LNB) [?] with differing parameter estimation settings; and (iii) the fully circular `IMRPhenomPv2NRT` family, employed by the LVK consortium.

Fig. ?? presents a comparative analysis of the posterior distributions for $\mathcal{M}$, q, and e_0 obtained from GW170817 parameter estimation using the `TaylorF2ecck` and `TaylorF2ecc` waveform approximants. Complementary to this, Fig. ?? illustrates degeneracies and correlations in the joint posterior distributions across the m–e_0 and η–e_0 planes. Our primary comparison in Fig. ?? focuses on non-spinning, 3PN-accurate eccentric waveform models, while Fig. ?? includes the 3.5PN version of `TaylorF2ecc`, with higher-order corrections applied only to the circular part of the phasing, to highlight potential biases introduced by such terms. Initial observations from Fig. ?? suggest that, at the negligible eccentricities inferred for GW170817, the 3PN `TaylorF2ecck` and 3PN `TaylorF2ecc` models yield broadly consistent parameter constraints. However, a notable bias emerges in the q posterior when comparing the 3PN `TaylorF2ecck` to the 3.5PN-corrected `TaylorF2ecc`, with the latter displaying a distinct shift in the posterior peak. This finding requires further explorations as eccentric contributions are fully known only upto the 3PN order.

For GW170817, all eccentric `TaylorF2` family waveforms favor essentially zero eccentricity, with 90% credible interval of $e_0 \le 0.009$ ($e_0 \le 0.006$) for `TaylorF2ecck`, using a log-uniform (uniform) prior of e_0, compared to $e_0 \le 0.005$ ($e_0 \le 0.004$) for `TaylorF2Ecc` (see Table ?? and Fig. ?? for more details on the parameter estimation results). In contrast, LNB while employing full-`TaylorF2Ecc` approximant reports 90% credible interval of $e_0 \le 0.024$ at 10 Hz using uniform prior, and their data shows $e_0 = 0.006^{+0.007}_{-0.005}$ at 20 Hz. The e_0 constraints obtained in our posterior results are notably lower than the 0.011 threshold identified through injection-recovery tests (see Fig. ??), and thus indicate negligible eccentricity. Moreover, we observed that inclusion of 3.5PN phasing in TaylorF2Ecc has negligible impact on the e_0 constraint as the implementation of eccentricity is limited to 3PN for this approximant. Furthermore, a Bayes factor comparison between TaylorF2ecck and TaylorF2Ecc shows no strong preference for either model (with $\Delta\log_{10}\mathrm{BF} < 1$), suggesting that the impact of eccentricity remains minimal in this scenario.

Fig. ?? provides posterior contour plot comparisons between the TaylorF2Ecck and TaylorF2Ecc in the m–e_0 and η–e_0 parameter spaces for the GW170817 event. The figure shows degeneracies between e_0 and the mass parameters, with an additional correlation visible in the higher e_0 region (> 0.002). A positive correlation is observed between m and e_0, while a negative correlation is observed between η and e_0. These correlations are consistent with results from our injection-recovery test with small eccentricity (see Fig. ?? in Appendix ?? for the posterior plot from the injection-recovery test with $e_0 = 0.01$ injection for a GW170817-like event). Moreover, Fig. ?? also displays distinct posterior distributions of the two waveform templates. Although the statistical uncertainty of e_0, indicated by the vertical width of (1, 2)-sigma contours in e_0 direction, exhibits a similar trend for both templates (i.e., increasing with increasing m), TaylorF2Ecck shows subtle differences from TaylorF2Ecc in the posterior distributions, as shown by the spread of (1, 2)-sigma contours, and this difference becomes more pronounced with increasing m. A similar behavior, where TaylorF2Ecck deviates from TaylorF2Ecc with increasing m, was observed in the match-test in e_0–m parameter space (see Fig. ?? of Sec. ??). However, given the negligible eccentricity constrained for this event, we don't expect significant differences in the posterior distributions of the two waveforms due to the inclusion of extra harmonics and periastron advance effects. This expectation is supported by the injection-recovery test in Sec. ??, which shows that TaylorF2Ecck becomes significantly distinguishable (with $\Delta\log_{10}\mathrm{BF} > 2$) from TaylorF2Ecc only at $e_0 > 0.06$.

Finally, the posterior distribution of the luminosity distance (D_L) relative to the inclination angle (θ_{jn}) shows remarkable similarity between TaylorF2ecck and TaylorF2Ecc, with %ΔMedian values less than 2% and $\mathcal{O}^{90\%}_{\mathrm{CI}}$ values above 99%. Unlike the mass parameters, no significant correlation appears between D_L and e_0, or θ_{jn} and e_0. Extended parameter posteriors are provided in Fig. ?? in the Appendix ?? for justifying these inferences.

A similar analysis of GW190425 reinforces the conclusions drawn from GW170817, although with greater uncertainty due to its lower SNR and lack of an electromagnetic counterpart. This leads to weaker constraints on the mass ratio and the eccentricity. While the inclusion of 3.5PN phasing in TaylorF2Ecc still induces a shift in the mass ratio, this effect is less pronounced than in GW170817. However, similar to GW170817, 3.5PN phasing in TaylorF2Ecc has a negligible impact on the e_0 constraint. The full parameter posteriors are shown in Fig. ?? in the Appendix ?? for additional reference.

For GW190425, TaylorF2Ecck and TaylorF2Ecc favor median e_0 values $\lesssim 0.011$, with upper 90% credible intervals of $e_0 < 0.02$ ($e_0 < 0.0413$) using a log-uniform (uniform) prior of e_0 (see Table ?? and Fig. ?? for more details on the parameter estimation results). These results are in line with prior of LNB, and their parameter estimation reports 90% credible interval with $e_0 \le 0.048$ ($e_0 \le 0.048$) at 10 Hz using a log-uniform (uniform) prior on e_0, and their data shows $e_0 = 0.00022^{+0.01601}_{-0.00022}$ ($e_0 = 0.013^{+0.012}_{-0.012}$) at 20 Hz using a log-uniform (uniform) prior of e_0, consistent with our result except for the trailing end and a small bump visible at slightly higher value of e_0 wrt the uniform prior. This is no such visible artifacts log-uniform prior (see Fig. ?? and Fig. ?? for full PE results wrt log-uniform and uniform prior respectively). That being said, for uniform prior since most volume of the e_0 posterior (1-sigma, $\tilde{6}8\%$) and for log-uniform prior 90% credible interval much lower than the 0.028 threshold identified through injection-recovery tests (see Fig. ?? in Appendix ??), this event can be considered to have negligible eccentricity. We did do extra parameter estimation just to be sure. Unlike GW170817, GW190429 doesn't suffer from loud glitch artifacts, but the main data we use (so does LNB) is the calibrated data with the glitch model subtracted. We repeated the parameter estimation with the calibrated data (but without the glitch subtraction). While the log-uniform prior results are consistent with our previously mentioned results, but for uniform prior, we didn't find the trailing and the bump of e_0 posterior, and this gives $e_0 < 0.028$ (see Fig. ??).

Similar to GW170817, the posterior distributions in the mass-eccentricity space (see m–e_0 and η–e_0 posterior contours in Fig. ??) show no significant difference between TaylorF2Ecck and TaylorF2Ecc due to this negligible eccentricity.

Furthermore, in our results, the Bayes factor difference between TaylorF2ecck and TaylorF2Ecc remains below 1, indicating no strong preference for one model over the other. The values of e_0 at 20 Hz and corresponding $\Delta\log_{10}\mathrm{BF}$ for the TaylorF2ecck and TaylorF2Ecc for the two inspiral events are listed in Table ??.

To further explore the observed mass ratio bias between the 3PN and the 3.5PN waveforms, we conducted additional checks using the circular TaylorF2 approximant with PN phasing terms. This extension seeks to clarify how higher PN corrections influence the precision and reliability of parameter estimation in eccentric binary coalescences, offering a pathway to refine our understanding of these subtle yet critical effects.

Event	e_0 Prior	Model	e_0 Value	$\Delta\log_{10}$BF
GW170817	Uniform	`Ecck`	$0.003^{+0.006}_{-0.002}$	368.41
		`Ecc`	$0.002^{+0.003}_{-0.002}$	368.43
	Log-Uniform	`Ecck`	$0.002^{+0.004}_{-0.001}$	369.23
		`Ecc`	$0.002^{+0.002}_{-0.001}$	369.26
GW190425	Uniform	`Ecck`	$0.011^{+0.030}_{-0.010}$	16.06
		`Ecc`	$0.011^{+0.030}_{-0.010}$	15.89
	Log-Uniform	`Ecck`	$0.004^{+0.016}_{-0.003}$	16.44
		`Ecc`	$0.004^{+0.015}_{-0.003}$	16.53

TABLE II. Parameter estimation result comparison of `TaylorEcck` (`Ecck`) and `TaylorEcck` (`Ecc`) waveforms for GW170817 and GW190425. It shows the values of medians and 90% credible intervals of e_0. Corresponding $\Delta\log_{10}$BF values are also included.

C. Probing PN order dependence of $\mathcal{M}$ and q posteriors for GW170817

The observed dependence of median q values on PN order up to which Fourier phase contributions are included, as evident from Fig. **??**, prompted us to explore if such q dependence exists for pure circular inspiral template families. In other words, does the quasi-circular `TaylorF2` approximant provide posteriors for GW170817 that depend on the PN accuracy of its Fourier phase contributions?

The `TaylorF2` approximant is typically used with 3.5PN accurate Fourier phase contributions, as done in the Ref. [**?**]. This approximant can be extended such that its Fourier phase is accurate to 4.5PN order [**?**]. This extension is primarily due to a highly detailed and demanding effort that extended GW generation to 4.5PN order by employing the Multipolar-Post-Minkowskian approach (see Ref. [**? ?**] and references therein). More importantly, the relevant Fourier coefficients at all PN orders up to 4.5PN were provided in [**?**]. Using these inputs, we extended the widely used `TaylorF2` approximant in LALSuite for parameter estimation to 4.5PN order, allowing us to use the template family with Fourier coefficients limited to 3PN, 3.5PN, 4PN, or 4.5PN order, which I will refer to as TF2-3PN, TF2-3.5PN, TF2-4PN, and TF2-4.5PN, respectively.

The results of our parameter estimation studies for the extended `TaylorF2` approximant are shown in Fig. **??**. Here, we set the sampling frequency to 2048 Hz with $f_{\max}$ of 512 Hz. Increasing these values should not alter our main conclusions. From Fig. **??**, it is clear that the median values for both $\mathcal{M}$ and q for TF2-3PN differ substantially from their counterparts at other PN order accuracies. Regarding q posterior distributions, the overlap between TF2-3PN and the other approximants is essentially zero, whereas the overlaps among TF2-3.5PN, TF2-4PN, and TF2-4.5PN are relatively high, with $\mathcal{O}^{90\%}_{\rm CI} \gtrsim$74%. In fact, $\mathcal{M}$ and q posteriors are relatively independent of PN orders from 3.5PN onwards. This is reassuring as our results indicate that the widely quoted $\mathcal{M}$ and q posteriors for GW170817 [**?**] are indeed correct.

A natural corollary of these parameter estimation studies is that it is desirable to extend the work of [**?**], which computed the angular momentum flux for eccentric binaries at 3PN order, to 3.5PN accuracy. This will involve computing 3.5PN accurate mass and current quadrupole moments that are critical to obtain 3.5PN accurate energy and angular momentum fluxes that involve mainly hereditary contractions [**? ?**]. After that, it should be straightforward to obtain $e_t(x, x_0, e_0)$ and $\Psi(x, x_0, e_0)$ to 3.5PN order following the prescription detailed in [**?**]. Interestingly, this extension would not require inputs from the 4PN accurate Keplerian type parametric solution (for eccentric, non-spinning binaries) available in [**?**], or the 4.5PN phasing if it becomes available later, to achieve reliable posterior results in parameter estimation.

In our opinion, it will be critical to obtain `TaylorF2ecck` approximant whose Fourier coefficients are accurate to 3.5PN order for data analysis purposes and while trying to obtain eccentric IMR template families that are required to search for eccentric GW events in the LVK data. Additionally, it will be interesting to explore the implications of missing 3.5PN order contributions to GW emissions in the time-domain eccentric IMR template families that are being constructed these days [**?**].

IV. CONCLUSION

In this work, we introduced and tested `TaylorF2ecck`, a TaylorF2 family frequency-domain inspiral approximant implemented within the LALSuite framework. This model, influenced by Ref. [**?**], analytically describes GWs from non-spinning compact binaries, incorporating eccentricity induced orbital, advance of periastron GW emission effects consistently to the 3PN order. Specifically, `TaylorF2ecck` incorporates $\mathcal{O}(e_0^2)$ contributions to the 3PN-accurate Fourier phases, while their amplitudes are at the quadrupolar order and with $\mathcal{O}(e_0)$ order accuracy.

Following comprehensive sanity checks to validate the model's performance and investigate the influence of periastron advance in the relevant parameter space, we employed `TaylorF2ecck` with the `Bilby` package to analyze public LVK data for the purely inspiral events GW170817 and GW190425. Our Bayesian parameter estimation indicates that the e_0 for both events is statistically consistent with zero at a 90% confidence level (specifically, $e_0 < 0.011$ for GW170817 and $e_0 < 0.028$ for GW190425 at 20 Hz), under the assumptions of uniform and log-uniform prior for e_0 and it corroborates previous studies by LNB. Given the negligible eccentricities measured for these events, our comparison between `TaylorF2ecck` and `TaylorF2ecc` showed subtle differences in the 2D mass-eccentricity posterior contours, but no significant differences in the constraints on individual parameters arising from the inclusion of multiple harmonics and the periastron advance effect in TaylorF2ecck. However, a substantial bias in the q posterior was observed when comparing these models to a version of TaylorF2ecc that included 3.5PN-order circular contributions in its Fourier phase. Considering the known correlation between mass and eccentricity, these parameter estimation studies therefore suggest the importance of incorporating e_0 contributions at least up to 3.5PN-order in our eccentric waveform model. This inference regarding the q bias and the significance of 3.5PN phasing was substantiated by employing versions of the quasi-circular TaylorF2 approximant that incorporate Fourier phase contributions at various PN orders.

A direct comparison can be made between LNB's result using full-TaylorF2Ecc and our result using TaylorF2Ecck regarding the constraints on e_0 for the GW190425 event. To give additional context to LNB's result, it is important to note the comparison with Ref. [?], which concerns the formation channel of GW190425. Specifically, Ref. [?] probes GW190425 for residual eccentricity at 10 Hz, to explore a formation channel influence by unstable BB mass transfer. Their result, with $e_0 \leq 0.007$, show no evidence for or against it. LNB reiterates this, reporting $e_0 \leq 0.023$ at 10 Hz, and states that this larger e_0 value doesn't alter the conclusion of Ref. [?]. Our result is consistent with LNB with $e_0 \leq 0.018$ (at 20 Hz) using log-uniform prior; additionally noting that we cannot distinguish eccentric waveforms from quasi-circular waveforms at low eccentricities.

Our results indicate that GW170817 and GW190425 show no evidence of dynamical formation due to their negligible e_0 values at 20 Hz, which is consistent with the binaries having formed in the field. This is supported by several reasons from prior studies: firstly, typical LVK binaries are expected to have circularized to $e_0 \leq 10^{-4}$ by the time they reach the LIGO-Virgo band [? ?]; secondly, only a small proportion (between 0.2% and 2%) of NS binaries will have eccentricity above 0.1 for the Advanced LIGO/Virgo [?]; and thirdly, the estimated merger rate of eccentric NS binaries formed via dynamical interactions in globular clusters is low ($\sim 0.02\mathrm{Gpc}^{-3}\mathrm{Yr}^{-1}$) [?]. However, it should be interesting to pursue an effort to constrain GW170817's e_0 value at 10 Hz using an improved version of TaylorF2Ecck while employing more efficient data analysis techniques (e.g., relative binning [?]). In contrast, such analyses will be more challenging for GW190425 due to its low network SNR, and this is reflected in the higher uncertainty in its e_0 estimates.

There are many directions to extend the present effort. It will be interesting to include higher order e_0 contributions both in Fourier phases and amplitudes of various harmonics, many of these contributions are available in Ref. [?], and explore their data analysis implications for both ground and space-based GW observatories [? ?]. Another direction should be to explore steps to create an efficient template bank for the detection of such binaries, influenced by Ref. [?]. It should be worthwhile to incorporate spin effects into our approximant and many required inputs exist in Ref. [? ?]. It will be desirable to add merger and ring-down effects into our pure inspiral templates by stitching our approximant with inputs from Numerical Relativity, as listed for example in Ref. [?]. Detailed parameter studies, relevant for both ground and space-based GW observatories should be another direction of our inspiral templates, influenced by Ref. [? ?]. However, all these efforts should employ TaylorF2Ecck approximant incorporates e_0 contributions appearing at 3.5PN order and this requires extending the detailed 3PN computations, detailed in Refs. [?], to 3.5PN order.

During the finalization of this manuscript, the pyEFPE waveform model became available [?], a new Python implementation of the long-developing EFPE framework [? ? ?] for modeling inspiraling compact binaries with eccentricity and spin-precession, which now also incorporates the effects of periastron advance. This model was recently used for parameter estimation of the GW200105 event (an NS-BH binary analyzed as a purely inspiral event by limiting $f_{\max}$ to 280 Hz), claiming evidence of periastron advance and reporting eccentricity of $e_0 = 0.145^{+0.007}_{-0.097}$ at 20 Hz[?]. Although our current study with TaylorF2Ecck primarily focuses on pure inspiral NS-NS events with generally low residual eccentricity ($\lesssim 0.1$), we performed a parameter estimation run on same event for comparison. A direct, one-to-one comparison between TaylorF2Ecck and pyEFPE is challenging due to differing methodologies to get GW waveform reponse function. TaylorF2Ecck, following Ref. [?], analytically implements eccentricity-induced orbital effects, including periastron advance and GW emission, and converts these to the frequency domain using SPA with fully analytic PN terms. In contrast, the pyEFPE model, as described in[?], incorporates eccentricity and periastron advance in a manner that does not allow for full analyticity in its frequency-domain eccentric part, due to numerical integration and other non-analytic approximations in handling eccentricity evolution. A preliminary mismatch comparison (Appendix ??) shows that pyEFPE diverges significantly from the TaylorF2 family for $e_0 \gtrsim 0.08$. This is likely due to TaylorF2Ecc/TaylorF2Ecck being limited to leading-order $\mathcal{O}(e_0^2)$ phasing corrections, whereas pyEFPE incorporates up-to $\mathcal{O}(e_0^4)$ terms, which are important for long inspirals and moderate e_0 (also see Ref. [?] for a comparison of $\mathcal{O}(e_0^2)$ and $\mathcal{O}(e_0^6)$ corrections in eccentric inspirals). Future updates to TaylorF2Ecck will enable direct tests of these higher-order corrections.

For the GW200105 event, using the same parameter estimation settings as in Ref. [?], our TaylorF2Ecck analysis yielded $e_0 = 0.122^{+0.027}_{-0.098}$ at 20 Hz, and an analysis with TaylorF2Ecc resulted in $e_0 = 0.122^{+0.028}_{-0.096}$. These results are consistent with those reported in[?] within the 90% credible intervals. Furthermore, parameter estimation runs using the circular TaylorF2 approximant with both 3PN and 3.5PN phasing corrections were conducted. The Bayes factor comparison yielded $\Delta \log_{10} \mathrm{BF} < 1$ when comparing `TaylorF2Ecck` to both `TaylorF2Ecc` and `TaylorF2` for this NS-BH event. Consequently, while our eccentricity estimates align, we cannot definitively claim the presence of periastron advance or eccentricity for this event with `TaylorF2Ecck` due to the lack of strong Bayesian evidence. This nevertheless suggests that such systems are promising candidates for future investigations, particularly with an improved version of `TaylorF2Ecc` approximant.

V. DATA AVAILABILITY

This article's data, code, and plot generation Jupyterlab-notebook are accessible in the associated data release on GitHub [?]. The version of LALSuite that includes `TaylorF2Ecck`, `TaylorF2Ecch`, and the extended PN-order `TaylorF2` is available in a forked GitLab repository [?] from the original repository [?].

For GW170817 and GW190425, we employ publicly available gravitational-wave strain data from the Advanced LIGO and Virgo detectors, accessed through the LIGO Open Science Center (LOSC) [?]. For GW170817, we utilize the LOSC_CLN_4_V1 data set, which includes noise subtraction performed by the collaboration [? ?]; additional details on the data release can be found in Ref. [?]. For GW190425, we use the T1700406_v3 data set, which features pre-processed glitch removal optimized for parameter estimation [? ?]; see also Ref. [?] for further information on the data release.

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Appendix A: Detailed Analytical Structure of the `TaylorF2Ecck` Waveform Approximant

The following subsections provide a comprehensive description of the analytical formulation of the `TaylorF2Ecck` waveform model. This includes the detailed mathematical expressions for the frequency-domain gravitational waveform $\tilde{h}(f)$, the associated Fourier phase $\psi_{j,n}$ for each harmonic (j, n), the periastron advance parameter k, and the formula for the evolution of orbital eccentricity e_t (following inputs from [?]).

1. The 3PN Accurate Analytical Expression for the Frequency-Domain Waveform $\tilde{h}(f)$

Below, we present the explicit of $\tilde{h}(f)$ with it's 3PN accurate $\psi_{j,n}$ with eccentricity correction of $\mathcal{O}(e_0^2)$ order, and quadrupolar amplitude with eccentricity correction of $\mathcal{O}(e_0)$ order. The expression reads

$$\tilde{h}(f) = \left(\frac{5\pi\eta}{384}\right)^{1/2} \frac{(Gm)^2}{c^5\mathcal{D}} \left(\frac{Gm\pi f}{c^3}\right)^{-7/6} \times \sum_{j,n} \xi_{j,n} \left(\frac{j}{2}\right)^{2/3} e^{-i(\Psi_{j,n}-\pi/4)}\Theta_{j,n}\,, \tag{A1a}$$

with

$$\begin{aligned}\Theta_{j,n} =&\,\text{Heaviside}\left[\left(j-(j+n)\frac{k}{1+k}\right)f_{\text{ISCO}} - 2f\right],\\ =&\begin{cases}1 & \text{if } \left(j-(j+n)\frac{k}{1+k}\right)f_{\text{LSO}} > 2f\,,\\ 0 & \text{if } \left(j-(j+n)\frac{k}{1+k}\right)f_{\text{LSO}} \le 2f\,,\end{cases}\end{aligned} \tag{A1b}$$

where f_{LSO} is the last stable circular orbit, and it reads

$$f_{\text{LSO}} = \frac{c^3}{Gm\pi 6^{3/2}}\,. \tag{A2}$$

2. Explicit 3PN Order Expression for the Fourier Phase $\psi_{j,n}$

The analytical form of 3PN-order Fourier phase of each (j,n) harmonic, $\psi_{j,n}$, is presented here. This phase is accurate to the 3PN order and includes terms that account for orbital eccentricity up to the order of $\mathcal{O}(e_0^2)$ contributions. The expression for $\psi_{j,n}$ reads

$$\psi_{j,n} = -2\pi f t_c + \phi_c\left(j-(j+n)\frac{k}{1+k}\right) - \frac{3j}{256\eta x_{j,n}^{5/2}}\sum_{u=0}^{6}\mathcal{D}_u\, x_{j,n}^{u/2}\,, \tag{A3a}$$

with

$$x_{j,n} = \left[\frac{2\pi Gmf}{\left\{j-\frac{(j+n)k}{1+k}\right\}c^3}\right]^{2/3}, \tag{A3b}$$

where the PN coefficients $\mathcal{D}_u$ in the above expression read

$$\mathcal{D}_0 = 1 - e_0^2\left\{\frac{2355}{1462}\chi^{-19/9}\right\}, \tag{A4a}$$

$$\mathcal{D}_1 = 0\,, \tag{A4b}$$

$$\mathcal{D}_2 = \frac{55\eta}{9} - \frac{25n}{3j} - \frac{2585}{756} + e_0^2\left\{\left(-\frac{128365\eta}{12432} + \frac{1805n}{172j} + \frac{69114725}{14968128}\right)\chi^{-19/9} + \left(\frac{154645\eta}{17544} - \frac{2223905}{491232}\right)\chi^{-25/9}\right\}, \tag{A4c}$$

$$\mathcal{D}_3 = -16\pi + e_0^2\left\{\frac{65561\pi}{4080}\chi^{-19/9} - \frac{295945\pi}{35088}\chi^{-28/9}\right\}, \tag{A4d}$$

$$\begin{aligned}\mathcal{D}_4 =&\, \frac{3085\eta^2}{72} + \eta\left(\frac{22105}{504} - \frac{10n}{j}\right) - \frac{31805n}{252j} - \frac{48825515}{508032}\\ &+ e_0^2\left\{\left(-\frac{10688155\eta^2}{294624} + \eta\left(\frac{36539875n}{1260072j} - \frac{72324815665}{6562454976}\right) + \frac{323580365n}{5040288j} + \frac{115250777195}{2045440512}\right)\chi^{-19/9}\right.\\ &+ \left(\frac{25287905\eta^2}{447552} + \eta\left(-\frac{355585n}{6192j} - \frac{3656612095}{67356576}\right) + \frac{5113565n}{173376j} + \frac{195802015925}{15087873024}\right)\chi^{-25/9}\\ &\left.+ \left(-\frac{14251675\eta^2}{631584} + \frac{3062285\eta}{260064} + \frac{936702035}{1485485568}\right)\chi^{-31/9}\right\},\end{aligned} \tag{A4e}$$

$$\begin{aligned}\mathcal{D}_5 =&\, -\left(\frac{65\pi\eta}{9} + \frac{160\pi n}{3j} + \frac{1675\pi}{756}\right)\left(\log\left(\frac{f}{f_{\text{LSO}}}\right) - \log(j)\right) - \frac{1}{9}(65\pi)\eta - \frac{32\pi n}{j} + \frac{14453\pi}{756}\\ &+ e_0^2\left\{\left(\frac{15803101\pi\eta}{229824} - \frac{4909969\pi n}{46512j} - \frac{458370775\pi}{6837264}\right)\chi^{-19/9}\right.\\ &+ \left(-\frac{48393605\pi\eta}{895104} + \frac{680485\pi n}{12384j} + \frac{26056251325\pi}{1077705216}\right)\chi^{-28/9}\\ &\left.+ \left(\frac{185734313\pi}{4112640} - \frac{12915517\pi\eta}{146880}\right)\chi^{-25/9} + \left(\frac{149064749\pi\eta}{2210544} - \frac{7063901\pi}{520128}\right)\chi^{-34/9}\right\},\end{aligned} \tag{A4f}$$

$$
\begin{aligned}
\mathcal{D}_6 =& -\frac{6848\gamma}{21} - \frac{127825\eta^3}{1296} + \eta^2\left(\frac{475n}{24j}+\frac{110255}{1728}\right) \\
&+ \eta\left(\frac{1845\pi^2 n}{32j} - \frac{2393105n}{1512j} + \frac{23575\pi^2}{96} - \frac{20562265315}{3048192}\right) + \frac{257982425n}{508032j} - \frac{3424\log(x)}{21} - \frac{640\pi^2}{3} \\
&+ \frac{13966988843531}{4694215680} - \frac{13696\log(2)}{21} + e_0^2\Bigg\{\Bigg(\frac{2105566535\eta^3}{10606464} + \eta^2\Bigg(-\frac{7198355375n}{45362592j} - \frac{2186530635995}{52499639808}\Bigg) \\
&+\eta\Bigg(-\frac{9519440485n}{35282016j} - \frac{13467050491570355}{39689727694848}\Bigg) + \frac{916703174045n}{5080610304j} + \frac{326505451793435}{2061804036096}\Bigg)\chi^{-25/9} \\
&+ \Bigg(-\frac{2330466575\eta^3}{16111872} + \eta^2\Bigg(\frac{32769775n}{222912j} + \frac{906325428545}{6466231296}\Bigg) \\
&+\eta\Bigg(-\frac{119702185n}{1560384j} - \frac{48415393035455}{1629490286592}\Bigg) - \frac{2153818055n}{524289024j} - \frac{82471214720975}{45625728024576}\Bigg)\chi^{-31/9} \\
&+ \Bigg(-\frac{734341\gamma}{16800} - \frac{69237581\eta^3}{746496} + \eta^2\Bigg(\frac{43766986495n}{1022720256j} - \frac{159596464273381}{1718170030080}\Bigg) \\
&+ \eta\Bigg(\frac{639805\pi^2 n}{22016j} - \frac{1219797059185n}{2045440512j} + \frac{12111605\pi^2}{264192} - \frac{37399145056383727}{28865256505344}\Bigg) + \frac{534109712725265n}{2405438042112j} \\
&- \frac{734341\log(x)}{33600} - \frac{21508213\pi^2}{276480} + \frac{4175723876720788380517}{5556561877278720000} + \frac{4602177\log(3)}{44800} - \frac{9663919\log(2)}{50400}\Bigg)\chi^{-19/9} \\
&+ \frac{24716497\pi^2}{293760}\chi^{-28/9} + \Bigg(\frac{2603845\gamma}{61404} + \frac{2425890995\eta^3}{68211072} + \frac{4499991305\eta^2}{636636672} \\
&+ \Bigg(\frac{3121945\pi^2}{561408} - \frac{1437364085977}{53477480448}\Bigg)\eta - \frac{2603845\log(\chi)}{184212} + \frac{2603845\log(x)}{122808} - \frac{96423905\pi^2}{5052672} - \frac{4165508390854487}{16471063977984} \\
&+ \frac{1898287\log(2)}{184212} + \frac{12246471\log(3)}{163744}\Bigg)\chi^{-37/9}\Bigg\}\,. \qquad \text{(A4g)}
\end{aligned}
$$

Additionally, we present the LALSuite-compatible form of $\psi_{j,n}$, where the reference phase ($\phi_{\rm ref}$), defined at a chosen reference frequency ($f_{\rm ref}$), is used instead of ϕ_c. If we define the third term of Eq. (**??**), i.e. "$\frac{3j}{256\eta x_{j,n}^{5/2}}\sum_{u=0}^{6}\mathcal{D}_u\, x_{j,n}^{u/2}$" as $\Psi_{j,n}$ (a function of frequency, provided fixed parameters) and given that $\Psi_{j,n}$ is $\Psi_{j,n}^{\rm ref}$ at $f_{\rm ref}$, then the form of $\psi_{j,n}$ with a proper phase shift reads

$$
\begin{aligned}
\psi_{j,n} =& -2\pi f t_c + \phi_{\rm ref}\left(j-(j+n)\frac{k}{1+k}\right) \\
&+ \Psi_{j,n}^{\rm ref} - \Psi_{j,n}\,. \qquad \text{(A5)}
\end{aligned}
$$

3. Explicit 3PN Order Expression for the Periastron Advance Parameter k

Complete expression of advancement of periastron parameter k is shown below and encompasses up-to 3PN terms and leading order in eccentricity $\mathcal{O}(e_0^2)$. k expression reads

$$
k_j = \sum_{u=0}^{6} k_u\, x_j^{u/2}\,, \qquad \text{(A6a)}
$$

with

$$
x_j = \left(\frac{2\pi G m f}{j c^3}\right)^{2/3}\,. \qquad \text{(A6b)}
$$

The coefficients of the above expression are listed as follows

$$
\begin{aligned}
k_0 =& \,0\,, \qquad \text{(A7a)} \\
k_1 =& \,0\,, \qquad \text{(A7b)}
\end{aligned}
$$

$$k_2 = 3 + e_0^2 \left\{ \frac{3}{\chi^{19/9}} \right\} , \tag{A7c}$$

$$k_3 = 0 , \tag{A7d}$$

$$k_4 = \frac{27}{2} - 7\eta + e_0^2 \left\{ \left(\frac{2833}{336} - \frac{197\eta}{12} \right) \frac{1}{\chi^{25/9}} + \left(\frac{10523}{336} - \frac{49\eta}{12} \right) \frac{1}{\chi^{19/9}} \right\} , \tag{A7e}$$

$$k_5 = e_0^2 \left\{ \frac{1}{\chi^{28/9}} + \frac{1}{\chi^{19/9}} \right\} \frac{377\pi}{24} , \tag{A7f}$$

$$\begin{aligned} k_6 = {} & \frac{1}{32}(2160 + (123\pi^2 - 5192)\eta + 224\eta^2) + e_0^2 \left\{ \left(-\frac{1193251}{1016064} - \frac{66317\eta}{3024} + \frac{18155\eta^2}{432} \right) \frac{1}{\chi^{31/9}} \right. \\ & \left. + \left(\frac{29811659}{338688} - \frac{276481\eta}{1512} + \frac{9653\eta^2}{432} \right) \frac{1}{\chi^{25/9}} + \left(\frac{142961921}{508032} - \frac{1419653\eta}{3024} + \frac{1599\pi^2\eta}{128} + \frac{91\eta^2}{27} \right) \frac{1}{\chi^{19/9}} \right\} . \end{aligned} \tag{A7g}$$

4. Explicit expression for eccentricity evolution e_t

Following Appendix C of [?], our 3PN accurate e_t expression that accounts for the leading order $\mathcal{O}(e_0)$ contributions reads

$$e_t = \sum_{m=0}^{6} \mathcal{E}_m \, x_{j,n}^{m/2}, \tag{A8}$$

with

$$\mathcal{E}_0 = e_0 \, \chi^{-19/18} , \tag{A9a}$$

$$\mathcal{E}_1 = 0 , \tag{A9b}$$

$$\mathcal{E}_2 = e_0 \left\{ \left(-\frac{2833}{2016} + \frac{197\eta}{72} \right) \chi^{-19/18} + \left(\frac{2833}{2016} - \frac{197\eta}{72} \right) \chi^{-31/18} \right\} , \tag{A9c}$$

$$\mathcal{E}_3 = e_0 \left\{ \frac{377}{144}\pi \left(-\chi^{-19/18} + \chi^{-37/18} \right) \right\} , \tag{A9d}$$

$$\begin{aligned} \mathcal{E}_4 = e_0 & \left\{ \left(\frac{77006005}{24385536} - \frac{1143767\eta}{145152} + \frac{43807\eta^2}{10368} \right) \chi^{-19/18} + \left(-\frac{8025889}{4064256} + \frac{558101\eta}{72576} - \frac{38809\eta^2}{5184} \right) \chi^{-31/18} \right. \\ & \left. + \left(-\frac{28850671}{24385536} + \frac{27565\eta}{145152} + \frac{33811\eta^2}{10368} \right) \chi^{-43/18} \right\} , \end{aligned} \tag{A9e}$$

$$\begin{aligned} \mathcal{E}_5 = e_0 & \left\{ \left(\frac{9901567\pi}{1451520} - \frac{202589\pi\eta}{362880} \right) \chi^{-19/18} + \left(-\frac{1068041\pi}{290304} + \frac{74269\pi\eta}{10368} \right) \chi^{-31/18} \right. \\ & \left. + \left(-\frac{1068041\pi}{290304} + \frac{74269\pi\eta}{10368} \right) \chi^{-37/18} + \left(\frac{778843\pi}{1451520} - \frac{4996241\pi\eta}{362880} \right) \chi^{-49/18} \right\} , \end{aligned} \tag{A9f}$$

$$\begin{aligned} \mathcal{E}_6 = & \left\{ \left(-\frac{33320661414619}{386266890240} + \frac{180721\pi^2}{41472} + \frac{3317\gamma}{252} + \left(\frac{161339510737}{8778792960} + \frac{3977\pi^2}{2304} \right) \eta - \frac{359037739\eta^2}{20901888} + \frac{10647791\eta^3}{2239488} \right. \right. \\ & + \frac{12091\log(2)}{3780} + \frac{26001\log(3)}{1120} + \left. \frac{3317\log(x)}{504} \right) \chi^{-19/18} + \left(\frac{218158012165}{49161240576} - \frac{346119344 51\eta}{1755758592} + \frac{191583143\eta^2}{6967296} \right. \\ & \left. - \frac{8629979\eta^3}{746496} \right) \chi^{-31/18} - \frac{142129\pi^2}{20736}\chi^{-37/18} + \left(\frac{81733950943}{49161240576} - \frac{6152132057\eta}{1755758592} - \frac{1348031\eta^2}{331776} \right. \\ & \left. + \frac{6660767\eta^3}{746496} \right) \chi^{-43/18} + \left(\frac{216750571931393}{2703868231680} + \frac{103537\pi^2}{41472} - \frac{3317\gamma}{252} + \left(\frac{866955547}{179159040} - \frac{3977\pi^2}{2304} \right) \eta \right. \\ & \left. \left. - \frac{130785737\eta^2}{20901888} - \frac{4740155\eta^3}{2239488} - \frac{12091\log(2)}{3780} - \frac{26001\log(3)}{1120} - \frac{3317\log(x)}{504} - \frac{3317\log(\chi)}{756} \right) \chi^{-55/18} \right\} e_0 . \end{aligned} \tag{A9g}$$

Appendix B: Detailed Analytical Structure of the `TaylorF2Ecch` Waveform Approximant

In the following subsections, we present the complete analytical structure of the `TaylorF2Ecch` waveform approximant. This includes explicit expressions for the frequency-domain waveform $\tilde{h}(f)$ and the associated Fourier phase ψ_j.

1. The 3PN Accurate Analytical Expression for the Frequency-Domain Waveform $\tilde{h}(f)$

Below, we present the explicit of $\tilde{h}(f)$ with it's 3PN accurate ψ_j with eccentricity correction of $\mathcal{O}(e_0^2)$ order, and quadrupolar amplitude with eccentricity correction of $\mathcal{O}(e_0)$ order. The expression reads

$$\tilde{h}(f) = \left(\frac{5\pi\eta}{384}\right)^{1/2} \frac{(Gm)^2}{c^5\mathcal{D}} \left(\frac{Gm\pi f}{c^3}\right)^{-7/6} \times \sum_j \xi_j \left(\frac{j}{2}\right)^{2/3} e^{-i(\Psi_j - \pi/4)} \Theta_j \,, \tag{B1}$$

where $\Theta_j = \text{Heaviside}\,[j f_{\text{LSO}} - 2f]$.

2. Explicit 3PN Order Expression for the Fourier Phase ψ_j

The analytical form of 3PN-order Fourier phase ψ_j, for each j-th harmonic, that includes $\mathcal{O}(e_0^2)$ contributions is given by

$$\psi_j = -2\pi f t_c + \phi_c j - \frac{3j}{256\eta x_j^{5/2}} \sum_{u=0}^{6} \mathcal{D}_u x_j^{u/2} \,, \tag{B2}$$

where $x_j = \left(\frac{2\pi G m f}{j c^3}\right)^{2/3}$. The PN coefficients $\mathcal{D}_u$ in the above expression read

$$\mathcal{D}_0 = 1 - e_0^2 \left\{ \frac{2355}{1462} \chi^{-19/9} \right\} , \tag{B3a}$$

$$\mathcal{D}_1 = 0 \,, \tag{B3b}$$

$$\mathcal{D}_2 = \frac{3715}{756} + \frac{55\eta}{9} + e_0^2 \left\{ \left(-\frac{2223905}{491232} + \frac{154645\eta}{17544} \right) \chi^{-25/9} + \left(-\frac{2045665}{348096} - \frac{128365\eta}{12432} \right) \chi^{-19/9} \right\} , \tag{B3c}$$

$$\mathcal{D}_3 = -16\pi + e_0^2 \left\{ -\frac{295945\pi}{35088} \chi^{-28/9} + \frac{65561\pi}{4080} \chi^{-19/9} \right\} , \tag{B3d}$$

$$\begin{aligned}
\mathcal{D}_4 = {} & \frac{15293365}{508032} + \frac{27145\eta}{504} + \frac{3085\eta^2}{72} \\
& + e_0^2 \left\{ \left(\frac{936702035}{1485485568} + \frac{3062285\eta}{260064} - \frac{14251675\eta^2}{631584} \right) \chi^{-31/9} + \left(-\frac{5795368945}{350880768} + \frac{4917245\eta}{1566432} + \frac{25287905\eta^2}{447552} \right) \chi^{-25/9} \right. \\
& \left. + \left(-\frac{111064865}{14141952} - \frac{165068815\eta}{4124736} - \frac{10688155\eta^2}{294624} \right) \chi^{-19/9} \right\} ,
\end{aligned} \tag{B3e}$$

$$\begin{aligned}
\mathcal{D}_5 = {} & \left(\frac{38645}{756} - \frac{65\eta}{9} \right) \pi \left(1 + \log\left(x^{3/2}\right) \right) + e_0^2 \left\{ \left(-\frac{7063901\pi}{520128} + \frac{149064749\pi\eta}{2210544} \right) \chi^{-34/9} \right. \\
& + \left(-\frac{771215705\pi}{25062912} - \frac{48393605\pi\eta}{895104} \right) \chi^{-28/9} + \left(\frac{185734313\pi}{4112640} - \frac{12915517\pi\eta}{146880} \right) \chi^{-25/9} \\
& \left. + \left(\frac{3873451\pi}{100548} + \frac{15803101\pi\eta}{229824} \right) \chi^{-19/9} \right\} ,
\end{aligned} \tag{B3f}$$

$$\begin{aligned}
\mathcal{D}_6 = {} & \frac{11583231236531}{4694215680} - \frac{640\pi^2}{3} - \frac{6848\gamma}{21} + \left(-\frac{15737765635}{3048192} + \frac{2255\pi^2}{12} \right) \eta + \frac{76055\eta^2}{1728} - \frac{127825\eta^3}{1296} - \frac{13696\log(2)}{21} \\
& - \frac{3424\log(x)}{21} + e_0^2 \left\{ \left(\frac{2440991806915}{1061063442432} + \frac{1781120054275\eta}{37895122944} - \frac{1029307085\eta^2}{150377472} - \frac{2330466575\eta^3}{16111872} \right) \chi^{-31/9} \right. \\
& + \frac{24716497\pi^2}{293760} \chi^{-28/9} + \left(-\frac{314646762545}{14255087616} - \frac{1733730575525\eta}{24946403328} + \frac{11585856665\eta^2}{98993664} + \frac{2105566535\eta^3}{10606464} \right) \chi^{-25/9} \\
& + \left(\frac{59648637301056877}{112661176320000} - \frac{21508213\pi^2}{276480} - \frac{734341\gamma}{16800} + \left(-\frac{409265200567}{585252864} + \frac{103115\pi^2}{6144} \right) \eta - \frac{4726688461\eta^2}{34836480} \right. \\
& \left. - \frac{69237581\eta^3}{746496} - \frac{9663919\log(2)}{50400} + \frac{4602177\log(3)}{44800} - \frac{734341\log(x)}{33600} \right) \chi^{-19/9} \\
& + \left(-\frac{4165508390854487}{16471063977984} - \frac{96423905\pi^2}{5052672} + \frac{2603845\gamma}{61404} + \left(-\frac{1437364085977}{53477480448} + \frac{3121945\pi^2}{561408} \right) \eta + \frac{4499991305\eta^2}{636636672} \right.
\end{aligned}$$

$$+\frac{2425890995\eta^3}{68211072}+\frac{1898287\log(2)}{184212}+\frac{12246471\log(3)}{163744}+\frac{2603845\log(x)}{122808}-\frac{2603845\log(\chi)}{184212}\Big)\chi^{-37/9}\Big\}\,. \tag{B3g}$$

Similar to Eq. (??), LALSuite-compatible form ψ_j can be written as follows

$$\psi_j = -2\pi f t_c + \phi_{\rm ref} j + \Psi_j^{\rm ref} - \Psi_j\,. \tag{B4}$$

Appendix C: Match-test in the Parameter Space of Eccentricity and Mass

This appendix section offers an extended analysis of the mismatch percentage, $\mathcal{M}^*_{\%}$, complementing comparisons in Fig. ?? of the main text. Fig. ?? details these comparisons, illustrating $\mathcal{M}^*_{\%}$ variations with m, e_0 (at 20 Hz), and η when comparing our `TaylorF2Ecck` approximant against `TaylorF2Ecc`, `TaylorF2Ecch`, and the circular `TaylorF2`. White dashed lines in the figure panels denote GW170817-like parameters. Notably, `TaylorF2Ecck` deviates more from `TaylorF2` at higher e_0 and lower m (fixed η), showing a positive correlation, and also at higher e_0 and lower η (fixed m), exhibiting a negative correlation. This deviation with m is more pronounced than with η. Similar behavior is anticipated when `TaylorF2Ecch` and `TaylorF2Ecc` are compared with `TaylorF2` due to their similarity in leading order eccentric correction in the dominant mode. Furthermore, `TaylorF2Ecck` increasingly deviates from `TaylorF2Ecch` with rising e_0 and m (fixed η), accompanied by a negative correlation. A similar, though more subtle, negative correlation is seen with increasing e_0 and η (fixed m); the change in $\mathcal{M}^*_{\%}$ with η in this comparison is less significant than with m, especially at lower e_0 values. The observed trends in the change of $\mathcal{M}^*_{\%}$ when comparing `TaylorF2Ecck` with `TaylorF2Ecch` are broadly similar to those seen in the `TaylorF2Ecck` versus `TaylorF2Ecc` comparison. This similarity can be attributed to the inclusion of the periastron advance effect in `TaylorF2Ecck`, which introduces extra harmonics and additional phase shifts in the sub-dominant harmonics compared to models that omit it.

Appendix D: Parameter Estimation Results from Injection-Recovery Tests

This appendix section presents supplementary results from injection-recovery tests, elaborating on the analysis in Section ??.

1. GW170817-like Injections

Fig. ?? shows two injection-recovery tests comparing `TaylorF2Ecck`, `TaylorF2Ecc`, and `TaylorF2`. It shows that `TaylorF2` fails to recover the injected mass parameters at $e_0 = 0.01$. Additionally, the plot illustrates how the constraints on the mass parameters and the correlation between them vary with changing e_0, using `TaylorF2Ecck` and `TaylorF2Ecc` waveforms.

2. GW190425-like Injections

Fig. ?? shows results from injection-recovery tests with a GW190425-like event and varying e_0 values. It compares `TaylorF2Ecck` with `TaylorF2` and illustrates the e_0 parameter region where `TaylorF2Ecck` becomes distinguishable from `TaylorF2`, and a value of e_0 where `TaylorF2` fails to recover the injected mass parameters.

Appendix E: Parameter Estimation Results of Pure Inspiral Events

We present extended posterior distributions for the binary neutron star events GW170817 and GW190425 using multiple waveform models to complement descriptions and plots in the main text.

1. Parameter estimation result for GW170817

Fig. ?? contextualizes `TaylorF2ecck`'s parameter estimation behavior against full-`TaylorF2ecc` (which includes leading-order spin and 3.5PN phase circular contributions) and LVK's `IMRPhenomPv2NRT (low-spin)`. We include two `TaylorF2ecc`related results: one derived from our parameter estimation and another from the prior work given in the Ref. [?], which we will refer to as LNB. LNB's results in Fig. ??, directly sourced from their data [?], should not be confused with the 10 Hz-associated findings presented in their paper. Notably, their work employs Markov Chain Monte Carlo (MCMC) parameter estimation with $f_{\rm min}$ of 20 Hz and inspiral duration of 190 seconds, which differs from ours (see Sec. ??).

In Fig. ??, we present the parameter estimation results for GW170817. These results include the posterior distributions of the following parameters: ($\mathcal{M}$, q, m, eta, e_0, D_L and θ_{jn}), and also show the correlations for each parameter pair. The plot compares `TaylorF2Ecck`, `TaylorF2Ecc`, and `TaylorF2Ecc` (the 3.5PN-accurate version), abbreviated as 'Ecck 3PN', 'Ecc 3PN', and 'Ecc 3.5PN', respectively. The Ecck 3PN and Ecc 3PN models produce nearly identical posteriors, with only subtle differences in the joint distributions involving e_0 (e.g., $\mathcal{M}$–e_0, q–e_0), attributed to periastron advance and higher harmonics in Ecck. A significant bias in the q posterior is observed when comparing Ecck 3PN to Ecc 3.5PN, reflecting the influence of higher-order circular corrections, though these corrections have negligible impact on the e_0 constraint. No significant differences are found in D_L or θ_{jn} posteriors, and neither correlates substantially with e_0.

2. Parameter estimation result for GW190425

Fig. **??** shows extended posterior distributions for GW190425 using the same waveform models and parameter set as in the GW170817 analysis. This analysis reinforces the conclusions drawn from GW170817, albeit with greater uncertainty due to GW190425's lower SNR and absence of an electromagnetic counterpart.

The inclusion of 3.5PN circular phasing in the Ecc model induces a shift in the q posterior, though this shift is less pronounced than in GW170817. Similar to GW170817, the e_0 constraint remains unaffected by the 3.5PN phasing. Subtle differences are again observed in the joint posteriors involving mass and eccentricity (e.g., $\mathcal{M}$–e_0, q–e_0). The D_L and θ_{jn} posteriors remain consistent across all waveform models and show no substantial correlation with e_0.

Appendix F: Comparison of `pyEFPE`, `TaylorF2Ecck`, and `TaylorF2Ecc` Waveform Approximants

Figure **??** presents a comparative analysis of the `pyEFPE` waveform model against members of the `TaylorF2` family of eccentric waveforms. Specifically, the mismatch percentage, $\mathcal{M}^*_{\%}$, is shown as a function of initial eccentricity e_0 for comparisons between `pyEFPE` and `TaylorF2Ecck` (at 3PN accuracy), as well as `TaylorF2Ecc` (at both 3PN and 3.5PN accuracy). The comparison is restricted to the inspiral regime by imposing upper frequency cutoffs of $f_{\rm max} = 512$ Hz and 1024 Hz, well below the last stable orbit frequency $f_{\rm LSO} \sim 1575$ Hz. This enables a meaningful inspiral-limited waveform comparison across different frequency ranges. Note that in other mismatch tests discussed in Sec. **??**, we used $f_{\rm max} = f_{\rm LSO}$.

A key observation from Figure **??** is that the mismatch $\mathcal{M}^*_{\%}$ between `pyEFPE` and the `TaylorF2` family waveforms does not approach zero even as $e_0 \to 0$, indicating fundamental differences between the models even in the quasi-circular limit. The mismatch also increases with higher $f_{\rm max}$, reflecting larger accumulated phase differences over extended inspiral durations. When comparing `pyEFPE` with `TaylorF2Ecc`, the mismatch is higher for the 3.5PN-accurate version than for the 3PN version, suggesting that including higher-order PN terms in `TaylorF2Ecc` increases its divergence from `pyEFPE`. In the comparison with `TaylorF2Ecck`, $\mathcal{M}^*_{\%}$ increases monotonically with e_0, crossing the $\mathcal{M}^*_{\%} = 3\%$ threshold at $e_0 \gtrsim 0.08$, implying significant mismatch of `pyEFPE` to `TaylorF2Ecck` for moderate eccentricities.

figs/amplitude_sanity_test_non_zero_e0_low_mass_BBH.pdf figs/mismatch_vs_eccn.pdf

(a) Match Test

figs/bf_vs_eccn.pdf

(b) Injection-Recovery Test

FIG. 2. Mismatch and Bayes factor comparisons for the `TaylorF2Ecck` waveform model relative to other inspiral approximants. Panel (a) shows the mismatch percentage ($\mathcal{M}^*_{\%}$) as a function of e_0, obtained from match tests comparing `TaylorF2Ecck` with `TaylorF2Ecch` (blue band), `TaylorF2Ecc` (orange band), and `TaylorF2` (green band). The colored bands represent results from multiple match calculations, varying the component masses m_1 and m_2 of GW170817 within the permitted binary neutron star prior ranges as outlined in Table 1 of Ref. [**?**]. The horizontal dashed line, corresponding to the $\mathcal{M}^*_{\%} = 3$ threshold, marks a 10% loss in matched-filtering signal power; its intersection points with the median of each band indicate the eccentricities at which waveform differences become statistically significant in matched-filtering searches. Consequently, for matched-filtering searches of BNS events, incorporating eccentricity becomes relevant at $e_0 \gtrsim 0.015$ (when comparing with `TaylorF2`), while accounting for periastron advance becomes important at $e_0 \gtrsim 0.16$ (vs. `TaylorF2Ecch`) and $e_0 \gtrsim 0.23$ (vs. `TaylorF2Ecc`). Panel (b) shows $\Delta \log_{10} \mathrm{BF}$ as a function of e_0 from injection-recovery analyses, comparing the `TaylorF2Ecck` model with `TaylorF2Ecch` (blue), `TaylorF2Ecc` (orange), and `TaylorF2` (green), using only the median component masses m_1 and m_2 of GW170817 due to computational constraints. The horizontal dashed lines at $\Delta \log_{10} \mathrm{BF} = 1$ and 2 represent thresholds for strong and decisive waveform distinguishability in parameter estimation, respectively[**?**]. The e_0 values at which the model comparison curves intersect the $\Delta \log_{10} \mathrm{BF} = 2$ line indicate when eccentricity and periastron advance become significant for parameter estimation of GW170817-like events. Following this, for such events, incorporating eccentricity becomes important at $e_0 \sim 0.011$ (when comparing with `TaylorF2`), and including periastron advance (when comparing with `TaylorF2Ecch` and `TaylorF2Ecc`) becomes important at $e_0 \gtrsim 0.06$.

figs/mismatchvalues_EcckVsEcc_fixed_q_EcckVsEcc.pdf figs/GW170817a.pdf

(a) Waveform comaprison plot

figs/GW170817c.pdf

(b) Waveform context plot

FIG. 4. Corner plots showing the posterior distributions of $\mathcal{M}$, q, and e_0 from parameter estimation for GW170817. The 2D contours represent the 1σ and 2σ credible intervals, enclosing 39.3% and 86.4% of the posterior samples, respectively. The left panel (a) compares two non-spinning, 3PN-accurate eccentric inspiral waveform models: `TaylorF2ecck` (Ecck 3PN) and `TaylorF2ecc` (Ecc 3PN). While this panel provides our primary comparison, the right panel (b) offers additional context by comparing `TaylorF2ecck` with the LALSuite-standard, but non-spinning, `TaylorF2ecc` (Ecc 3.5PN), which includes 3.5PN-order contributions to the circular part of the Fourier phase. In the left panel, both 3PN models constrain the residual eccentricity to be negligible ($e_0 \lesssim 0.01$). At these low eccentricities ($\lesssim 0.01$), the comparison between the Ecck 3PN and Ecc 3PN models reveals subtle differences in the joint posteriors involving mass and eccentricity (e.g., $\mathcal{M}$–e_0 and q–e_0 planes), while the 1D marginalized distributions show no significant variation due to the inclusion of additional harmonics or periastron advance in `TaylorF2ecck`. However, the right panel shows a substantial bias in the q posterior when comparing the 3PN models to the 3.5PN `TaylorF2ecc`, highlighting the influence of higher-order circular phase corrections. Notably, the inclusion of 3.5PN circular phasing in `TaylorF2Ecc` has a negligible impact on its e_0 constraint, as the eccentricity implementation in this approximant is limited to 3PN order.

figs/GW170817_mtot_eta_e0_edited.pdf

FIG. 5. Plot shows contour plots for m vs e_0 (left) and η vs e_0 (right), while comparing the waveform approximants `TaylorF2ecck` (Ecck, in blue) and `TaylorF2ecc` (Ecc, in orange). The plot shows posterior distribution accociated with GW170817. Contour lines indicates (1, 2)-sigma or 39.3% and 86.4% of the posterior distribution. The plot shows positive correlation for m vs e_0 and negative correlation for η vs e_0. Ecck deviates more from Ecc at higher value of m indicating the dependence of periastron effect on PN parameter.

figs/GW170817_TF2.pdf

FIG. 6. Chirp mass $\mathcal{M}$ and mass ratio q posteriors for GW170817 derived from the circular `TaylorF2` approximant (TF2) with varying PN order accuracy (3PN, 3.5PN, 4PN and 4.5PN) in the Fourier phase. Despite some residual uncertainties inherent in parameter estimation, these posteriors largely converge beyond 3PN, considering the median values and the 90% credible intervals. The natural implication of these parameter estimation studies is that such inspiral approximants, including the `TaylorF2Ecck` and `TaylorF2Ecc`, should be at least 3.5PN accurate in their GW phase evolution to get a reliable posterior.

figs/mismatchvalues_EcckVsTF2_fixed_q.pdf figs/injection_recovery_Ecck_vs_Ecc_e0_0p01_GW170817.pdf

(a) Injection-recovery with $e_0 = 0.01$

figs/injection_recovery_Ecck_vs_Ecc_e0_0p10_GW170817.pdf

(b) Injection-recovery with $e_0 = 0.1$

FIG. 8. Corner plots illustrating the posterior distributions of m, η, and e_0 from injection-recovery tests using GW170817-like parameters with non-spinning assumptions. These tests evaluate the performance of three waveform approximants: `TaylorF2Ecck` (Ecck 3PN), `TaylorF2Ecc` (Ecc 3PN), and `TaylorF2` (TF2 3PN, left panel only) in recovering signals injected with `TaylorF2Ecck` at two initial eccentricity values, $e_0 = 0.01$ (panel a) and $e_0 = 0.1$ (panel b), both at 20 Hz. In each panel, the vertical solid black lines indicate the injected values, while the vertical dotted lines show the median and 90% credible intervals for each model: Ecck (blue), Ecc (orange), and TF2 (red). The 2D contours represent 39.3% and 86.4% credible regions. At $e_0 = 0.01$, Ecck and Ecc both successfully recover the injected mass parameters, while TF2 fails to do so within the 90% interval. At $e_0 = 0.1$, Ecck not only provides tighter parameter constraints but also displays an altered correlation structure between mass and eccentricity, highlighting the increasing importance of including eccentricity and periastron advance at higher e_0. See Sec. **??** for additional discussion.

figs/bf_vs_eccn_tf2_GW190425.pdf

(a) $\Delta\log_{10}$BF Vs e_0 plot for GW190425-like event

figs/injection_recovery_Ecck_vs_TF2_e0_0p05_GW190425.pdf

(b) Injection-recovery result plot (with $e_0 = 0.02$) for GW190425-like event

FIG. 9. Injection-recovery analysis for events, with GW190425-like parameters, using eccentric waveform models. The aim is to assess the waveform distinguishability between `TaylorF2Ecck` (Ecck, 3PN-accurate, non-spinning) and the quasi-circular `TaylorF2` (TF2, 3PN-accurate, non-spinning) across a range of initial eccentricities e_0. Panel (a) shows $\Delta \log_{10}$ BF as a function of e_0, illustrating when the inclusion of eccentricity becomes statistically significant. Horizontal dashed lines mark conventional interpretative thresholds: $\Delta \log_{10} \mathrm{BF} = 1$ (strong), 2 (decisive), and 8 (Jeffreys' criterion for strong evidence). Vertical lines correspond to eccentricity values ($e_0 = 0.018$ and $e_0 = 0.028$) beyond which `TaylorF2Ecck` becomes distinguishable from TF2 under different criteria. Panel (b) presents the posterior distributions of chirp mass $\mathcal{M}$ and mass ratio q for an injected signal with $e_0 = 0.02$ using `TaylorF2Ecck`. The model recoveries from Ecck (blue) and TF2 (red) are shown with median values and 90% credible intervals as vertical dashed lines; the injected values are indicated by solid black lines. The plot demonstrates that TF2 fails to recover the injected mass parameters within the 90% interval, even at this low e_0, while Ecck succeeds. This highlights the importance of accounting for eccentricity in waveform models when analyzing such systems.

figs/GW170817b.pdf

FIG. 10. Corner plots showing the posterior distributions for $\mathcal{M}$, q, and e_0 from parameter estimation of GW170817. The 2D contours represent the 1σ and 2σ credible intervals, enclosing 39.3% and 86.4% of the posterior samples. This comparison includes results from our 3PN-accurate `TaylorF2ecck` approximant (Ecck 3PN) against those arising from the standard full-`TaylorF2ecc` approximant that incorporates the leading order spin effects and 3.5PN contributions for circular orbits in its Fourier phase from i) our parameter estimation results (Ecc 3.5PN+Spin) and ii) Lenon et al. 2020 [**?**] (Ecc 3.5PN+Spin LNB) with differing parameter estimation settings, and iii) the fully circular `IMRPhenomPv2NRT` family (Pv2NRT), employed by the LVK consortium. The Ecck model provides tighter constraints on q than the spin-corrected models, though this must be interpreted cautiously as Ecck lacks spin effects and higher-order (3.5PN) phasing corrections. The Ecc model in our analysis yields tighter posteriors compared to LNB, likely due to differences in sampling techniques (Nested Sampling vs. MCMC) and the inclusion of longer-duration data. The Pv2NRT model is included for broader context and comparison.

figs/fullpe_gw170817.pdf

FIG. 11. Figure shows extended corner plots of the posterior results from the GW170817 parameter estimation, complementary to Fig. **??**. The 2D contour plots indicate the 1σ and 2σ confidence regions, corresponding to 39.3% and 86.4% of the posterior samples, respectively. Vertical dashed lines mark the median and 90% credible intervals of the `TaylorF2Ecck` posteriors. Three waveform models are compared: Ecck (3PN), `TaylorF2Ecc` (Ecc 3PN), and `TaylorF2Ecc` with additional 3.5PN circular corrections (Ecc 3.5PN). The comparison focuses on key binary parameters: $\mathcal{M}$, q, m, η, e_0, D_L, and θ_{jn}. Given the observed negligible eccentricity (i.e., $e_0 \lesssim 0.011$), the Ecck 3PN and Ecc 3PN models yield very similar posterior distributions for most parameters, with only subtle differences observed in the mass and eccentricity joint posteriors (e.g., $\mathcal{M}$–e_0 and q–e_0 planes), due to the inclusion of periastron advance and additional harmonics in Ecck 3PN. However, a significant bias is seen in the q posterior when comparing Ecck 3PN to Ecc 3.5PN, underscoring the impact of higher-order circular phasing terms. Notably, these 3.5PN circular corrections in Ecc 3.5PN do not affect its e_0 constraint, as the eccentricity implementation in this approximant is limited to 3PN order. The posteriors for D_L and θ_{jn} show no significant differences across the models and exhibit no substantial correlation with e_0.

figs/fullpe_gw190425.pdf

FIG. 12. Figure displays extended corner plots of the posterior results from the GW190425 parameter estimation analysis with $f_{\rm min}$ set at 20 Hz. The 2D contour plots indicate the 1σ and 2σ confidence regions, corresponding to 39.3% and 86.4% of the posterior samples, respectively. Vertical dashed lines mark the median and 90% credible intervals of the `TaylorF2Ecck` posteriors. Three waveform models are compared: Ecck (3PN), `TaylorF2Ecc` (Ecc 3PN), and `TaylorF2Ecc` with additional 3.5PN circular corrections (Ecc 3.5PN). The parameters analyzed include $\mathcal{M}$, q, m, η, e_0, D_L, and θ_{jn}. GW190425 reinforces the conclusions drawn from GW170817, although with greater uncertainty due to its lower signal-to-noise ratio and the absence of an electromagnetic counterpart. This results in weaker constraints on both the mass ratio q and the eccentricity e_0, with e_0 remaining consistent with zero (i.e., $e_0 \lesssim 0.028$). While the inclusion of 3.5PN circular phasing in `TaylorF2Ecc` still induces a shift in the mass ratio posterior, the effect is less pronounced than for GW170817. Similar to GW170817, subtle differences are observed in the joint posteriors involving mass and eccentricity (e.g., $\mathcal{M}$–e_0 and q–e_0 planes). The inclusion of 3.5PN phasing in `TaylorF2Ecc` has a negligible impact on the e_0 constraint. Furthermore, the posteriors for D_L and θ_{jn} show no significant differences across waveform models and exhibit no substantial correlation with e_0.

figs/mismatch_pyEFPE_fmax_512Hz.pdf

(a) $f_{\rm max} = 512$ Hz

figs/mismatch_pyEFPE_fmax_1024Hz.pdf

(b) $f_{\rm max} = 1024$ Hz

FIG. 13. Mismatch comparison between `pyEFPE` and TaylorF2-family eccentric waveform approximants: `TaylorF2Ecck` (3PN), `TaylorF2Ecc` (3PN), and `TaylorF2Ecc` (3.5PN). The mismatch $\mathcal{M}^*_{\%}$ is plotted as a function of e_0 using a GW170817-like (non-spinning) binary and inspiral-only comparison for two different frequency cutoffs: (a) $f_{\rm max} = 512$ Hz and (b) $f_{\rm max} = 1024$ Hz. Across all cases, `pyEFPE` shows non-negligible mismatch with the TaylorF2 approximants, with $\mathcal{M}^*_{\%}$ increasing at higher e_0 and $f_{\rm max}$. Notably, mismatch with `TaylorF2Ecck` exceeds 3% for $e_0 \gtrsim 0.08$, pointing to significant structural differences, possibly due to non-analytic eccentricity treatment and harmonic content in `pyEFPE`.